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(19) **United States**(12) **Patent Application Publication** (10) **Pub. No.: US 2025/0283028 A1****Jauert et al.**(43) **Pub. Date: Sep. 11, 2025**(54) **GENETICALLY MODIFIED YEAST AND FERMENTATION PROCESSES FOR THE PRODUCTION OF XYLITOL**(71) Applicant: **Cargill, Incorporated**, Wayzata, MN (US)(72) Inventors: **Peter Alan Jauert**, Minneapolis, MN (US); **Nandita Kohli**, Plymouth, MN (US); **Christopher Kenneth Miller**, Andover, MN (US); **Gregory Michael Poynter**, Saint Paul, MN (US); **Maria Isabel Sardi**, Plymouth, MN (US)(73) Assignee: **Cargill, Incorporated**, Wayzata, MN (US)(21) Appl. No.: **18/863,697**(22) PCT Filed: **May 5, 2023**(86) PCT No.: **PCT/US2023/066627**

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(57)

ABSTRACT

Disclosed herein are genetically engineered yeast cells capable of producing xylitol. The engineered yeast cell may comprise an exogenous polynucleotide sequence encoding a sugar phosphatase enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs: 8 and 20.

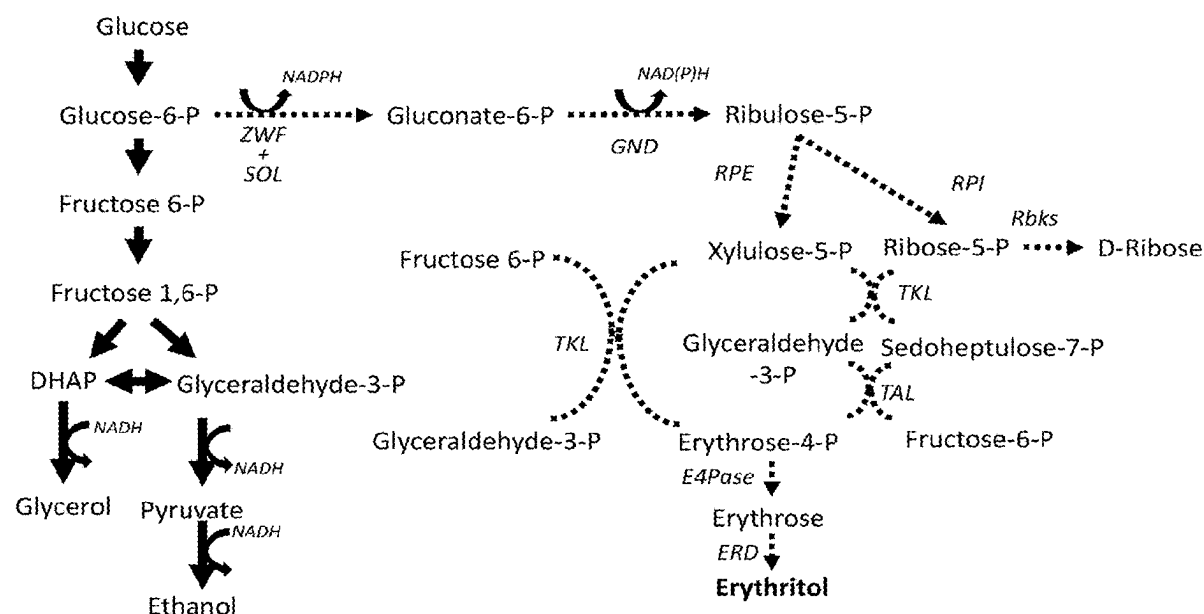
Specification includes a Sequence Listing.

FIG. 1

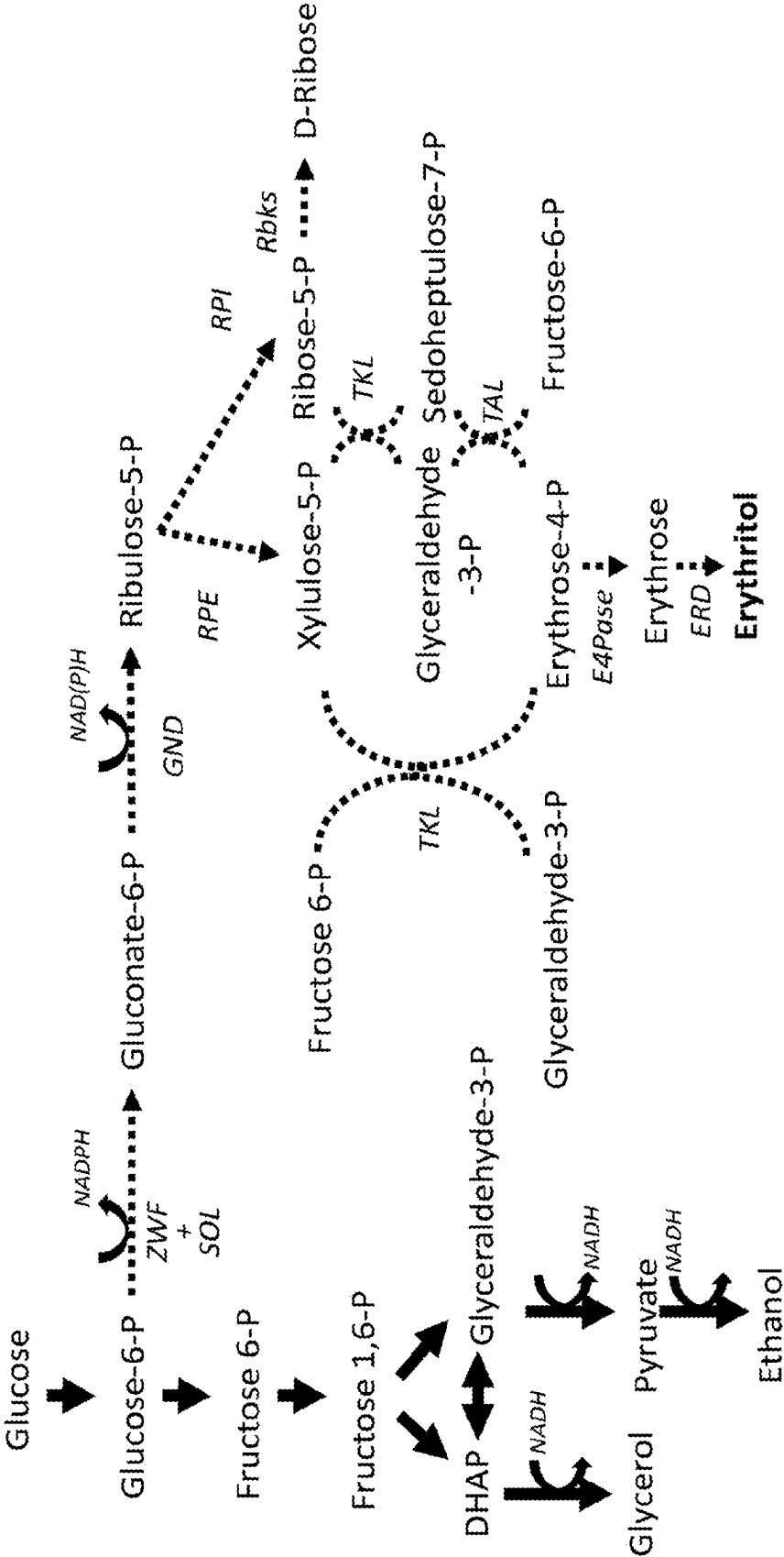


FIG. 2

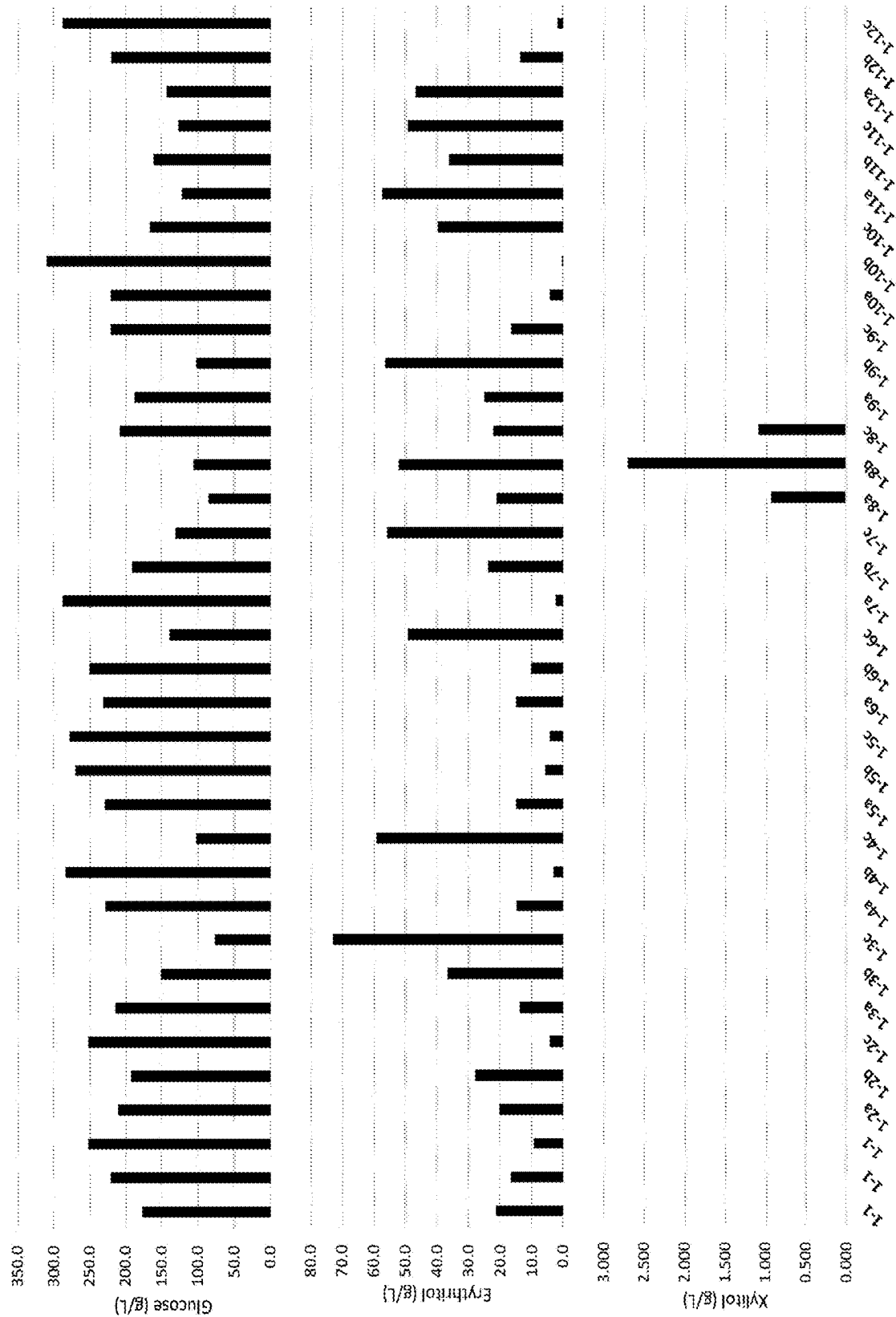


FIG. 3

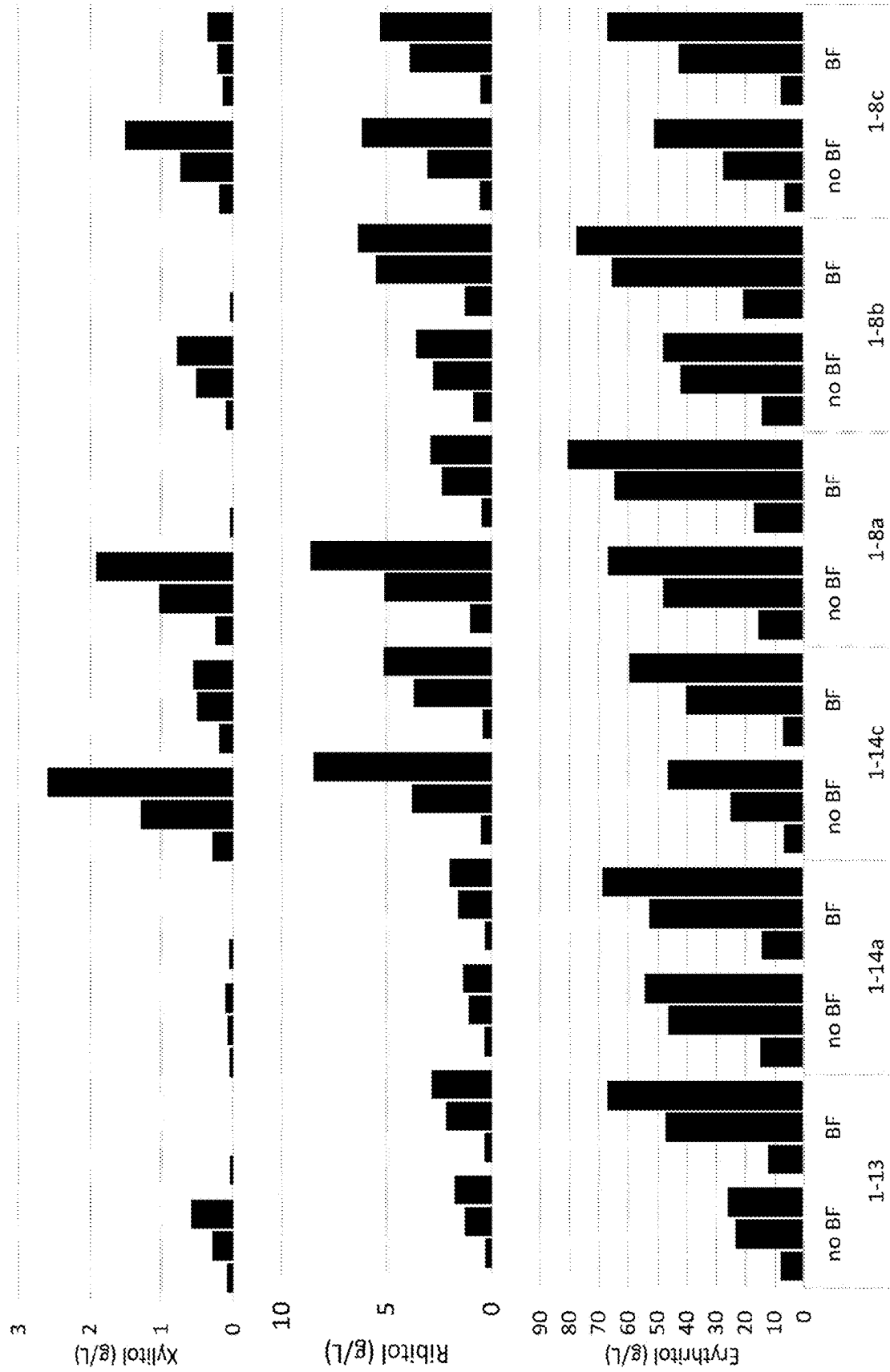


FIG. 4

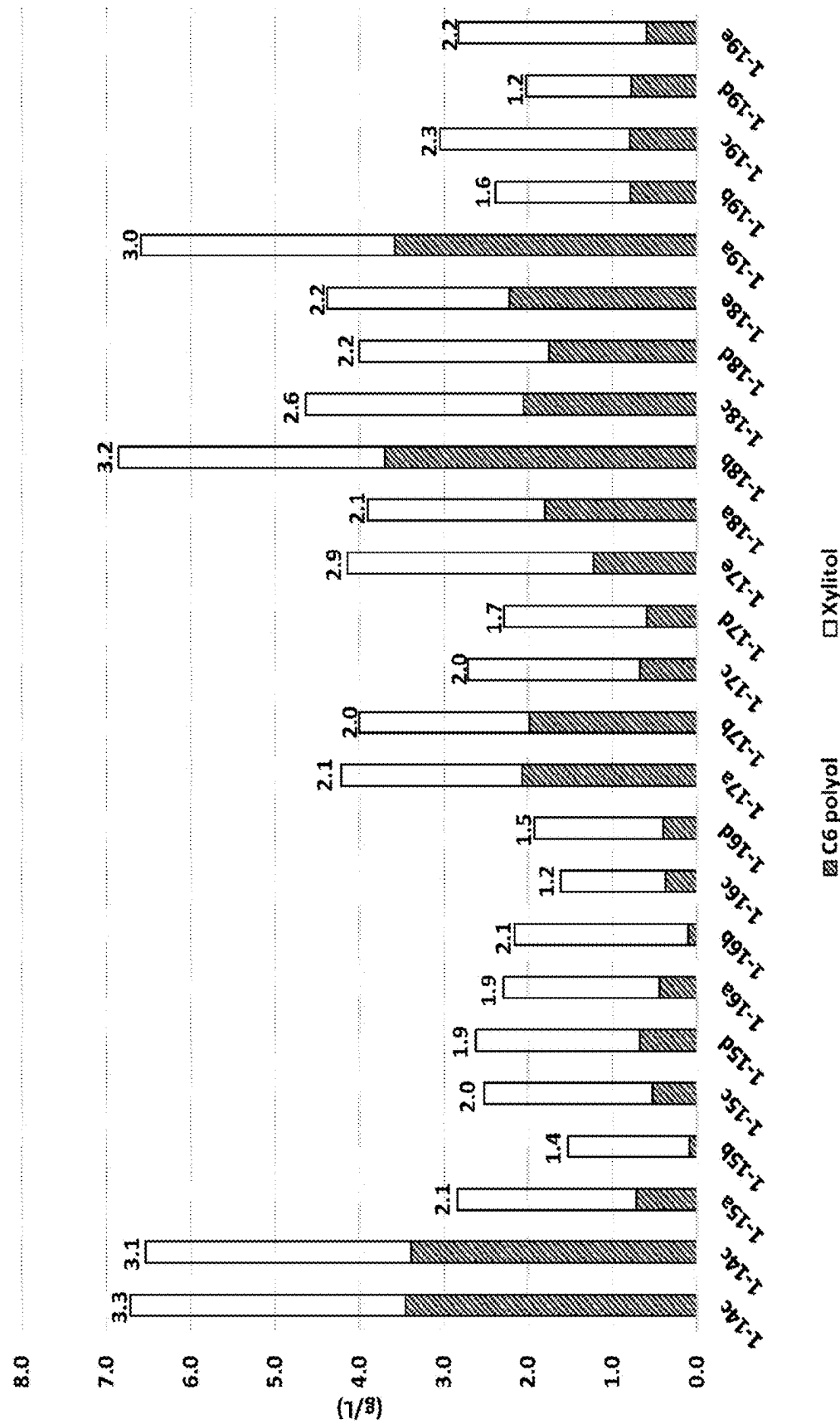


FIG. 5

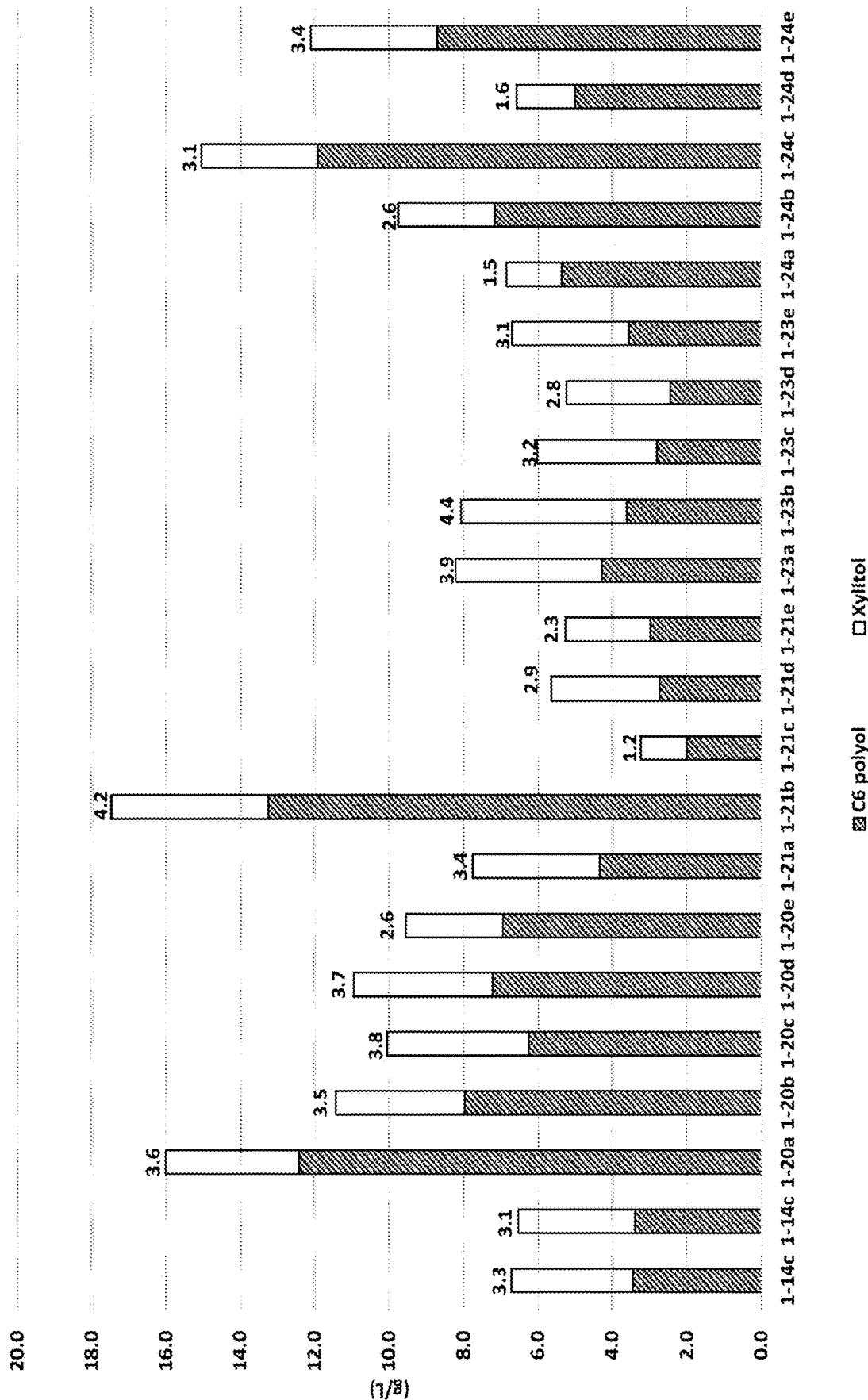
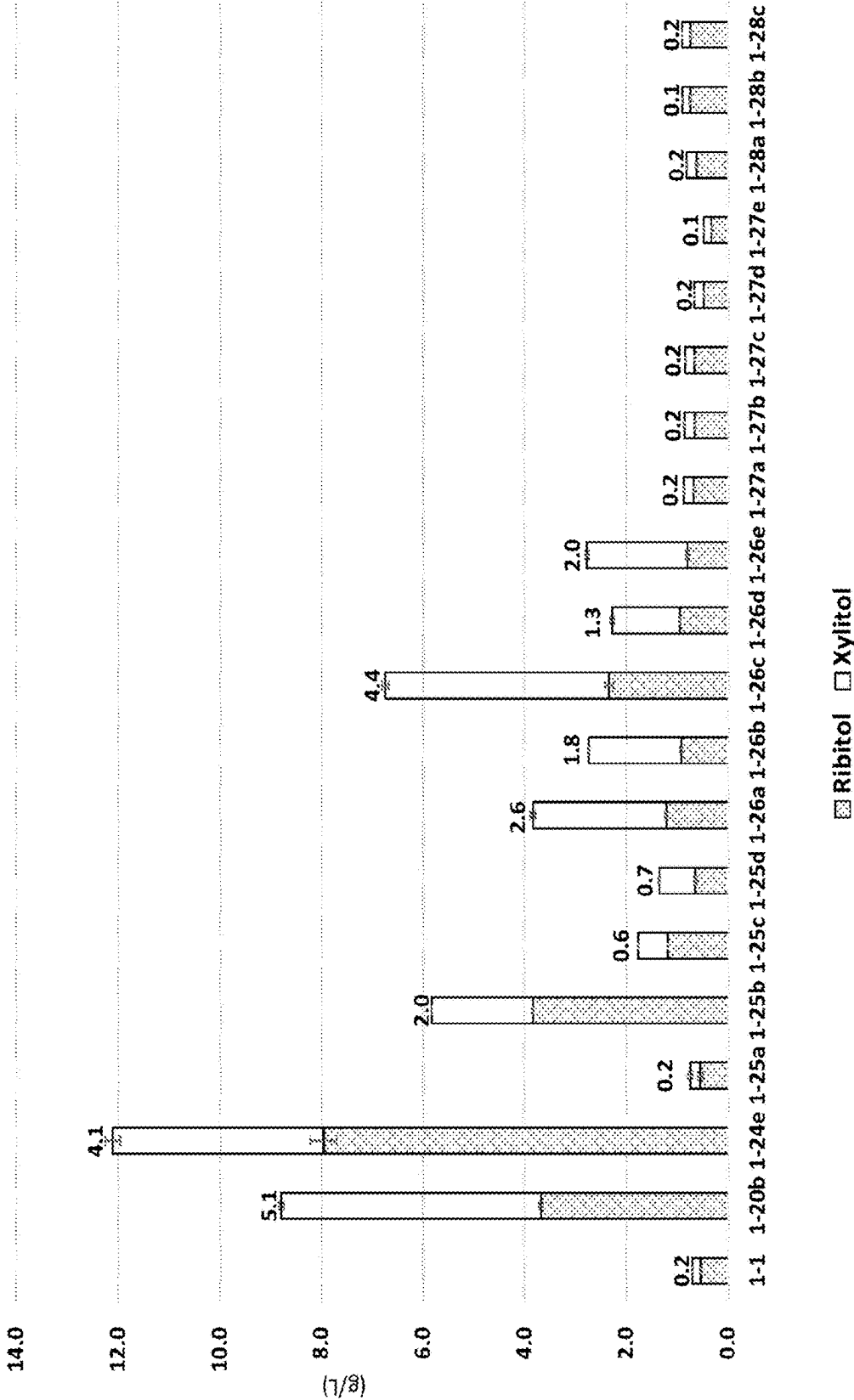


FIG. 6



GENETICALLY MODIFIED YEAST AND FERMENTATION PROCESSES FOR THE PRODUCTION OF XYLITOL

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/364,363, filed May 9, 2022, which is incorporated by reference herein in its entirety.

REFERENCE TO A SEQUENCE LISTING SUBMITTED VIA PATENT CENTER

[0002] The content of the Sequence Listing XML file of the sequence listing named "PT-1346-WO-PCT.xml" which is 246,439 bytes in size created on Apr. 13, 2023 and electronically submitted via Patent Center herewith the application is incorporated by reference in its entirety.

BACKGROUND

[0003] Xylitol is a low-calorie sweetener used as a food additive and sugar substitute. Commonly used in drug, dietary supplement, confectionary, and toothpaste compositions, xylitol has also been associated with anticariogenic properties when used in chewing gums. Traditional methods of xylitol production, including chemically catalyzed hydrogenation of xylose hydrolyzed from biomass extracted xylan, are both monetarily and environmentally costly. These methods require high temperatures and pressures, large amounts of water, and metal catalysts that must be mined. In contrast, fermentation processes have been used commercially at large scale to produce other organic molecules, such as ethanol, citric acid, lactic acid, and the like, and may offer a cost effective and sustainable alternative to traditional xylitol processing methods.

[0004] Accordingly, provided herein are genetically modified yeast and fermentation methods for the production of xylitol while reducing production of erythritol.

SUMMARY

[0005] The present disclosure provides a genetically engineered yeast cell capable of producing xylitol, the engineered yeast cell comprising an exogenous polynucleotide sequence encoding a xylulokinase (XKS) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs:8 and 20. The yeast cell may be an osmotolerant yeast cell. The yeast cell may be a cell of the subphylum Ustilaginomycotina. The yeast cell may be selected from the group consisting of *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*, *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Moniliella*, *Ustilaginomycetes*, *Trichosporon*, *Yarrowia lipolytica*, *Penicillium*, *Torula*, *Pichia*, *Candida*, *Candida magnoliae*, and *Aureobasidium*. The yeast cell may be a yeast cell of the genus *Moniliella*.

[0006] The disclosure also provides a genetically engineered *Moniliella* cell capable of producing xylitol, the engineered *Moniliella* cell comprising an exogenous polynucleotide sequence encoding a xylulokinase (XKS) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least

90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs:8 and 20.

[0007] The XKS enzyme may have a sequence at least 85% identical to at least one of SEQ ID NOs:8 and 20. The XKS enzyme may have a sequence at least 90% identical to at least one of SEQ ID NOs:8 and 20.

[0008] The engineered cell described herein may be a *Moniliella pollinis* cell. The yeast cell described herein may be capable of producing xylitol at a titer of at least 0.5, 1.0, 1.5, 2.0, or 2.5 g/L when used in a fermentation process in the presence of dextrose for 96 hours. Erythritol production by the engineered cell described herein may be reduced relative to erythritol production in an equivalent yeast cell lacking the exogenous polynucleotide sequence.

[0009] The exogenous polynucleotide sequence may be operably linked to a heterologous or artificial promoter. The promoter may be a constitutive promoter. The promoter may be selected from the group consisting of pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:61), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:52), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:54), translational elongation factor 1 promoter (TEFp; SEQ ID NO:55), modified TEFp (SEQ ID NO:53), phosphoglucosyltransferase 1 promoter (PGM1p; SEQ ID NO:56), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:57), enolase promoter (ENO1p; SEQ ID NO:58), asparagine synthetase promoter (ASNSp; SEQ ID NO:59), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:60), and RPL16B (SEQ ID NO:62).

[0010] The exogenous polynucleotide sequence may be integrated into the genome of the yeast cell at a locus selected from the ER1 locus, the ER3 locus, the PDC1 locus, the pyrF locus, the TRP3 locus, the gpdIIA locus, and the gpdIIB locus.

[0011] The yeast cell may additionally comprise an exogenous polynucleotide sequence encoding a xylitol dehydrogenase (XDH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs:1, 13, and 14. The XDH enzyme may have a sequence at least 85% identical to at least one of SEQ ID NOs:1, 13, and 14. The XDH enzyme may have a sequence at least 90% identical to at least one of SEQ ID NOs:1, 13, and 14.

[0012] The disclosure also provides a method for producing xylitol using the engineered cells described herein, the method comprising contacting a substrate comprising dextrose with an engineered cell described herein, wherein fermentation of the substrate by the engineered yeast produces xylitol. The disclosure also provides a method for producing xylitol, the method comprising contacting a substrate comprising dextrose with an engineered yeast cell comprising an exogenous polynucleotide sequence encoding a sugar phosphatase enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs:8 and 20, wherein fermentation of the substrate by the engineered yeast produces xylitol. The fermentation temperature may be at or between 25° C. to 45° C., 30° C. to 40° C., or 32° C. to 37° C. The volumetric oxygen uptake rate (OUR) may be between 0.5 to 40, 1 to 35, 2 to 30, 3 to 25, 4 to 20, or 5 to 15 mmol O₂/(L·h). Erythritol production may be reduced relative to an equivalent fermentation run with an

equivalent yeast cell lacking the exogenous polynucleotide sequence. Erythritol production may be less than 40, 30, or less than 20 g/L when the fermentation is run at 35° C. for 96 hours. Xylitol production may be at least 0.5, 1.0, 1.5, 2.0, 2.5, or 3.0 g/L when the fermentation is run at 35° C. for 96 hours. Glycerol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence. Ethanol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

BRIEF DESCRIPTION OF THE FIGURES

[0013] The drawings illustrate generally, by way of example, but not by way of limitation, various aspects discussed herein.

[0014] FIG. 1 shows the native pentose phosphate pathway (dotted lines and arrows) and the native glycolysis pathways (solid lines and arrows) in *Moniliella pollinis*.

[0015] FIG. 2 shows glucose, erythritol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-1, 1-2a-c, 1-3a-c, 1-4a-c, 1-5a-c, 1-6a-c, 1-7a-c, 1-8a-c, 1-9a-c, 1-10a-c, 1-11a-c, and 1-12a-c as outlined in Example 2.

[0016] FIG. 3 shows xylitol, ribitol, and erythritol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-13, 1-14a, 1-14c, 1-8a, 1-8b, and 1-8c both with baffled flasks (BF) and without baffled flasks (no BF), as outlined in Example 3.

[0017] FIG. 4 shows C6 polyol and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-14c, 1-15a-d, 1-16a-d, 1-17a-d, 1-18a-e, and 1-19a-e, as outlined in Example 4. Data labels report the concentration (g/L) of xylitol.

[0018] FIG. 5 shows C6 polyol and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-14c, 1-20a-e, 1-21a-e, 1-23a-e, and 1-24a-e, as outlined in Example 5. Data labels report the concentration (g/L) of xylitol.

[0019] FIG. 6 shows ribitol and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-1, 1-20b, 1-24e, 1-25a-d, 1-26a-e, 1-27a-e, and 1-28a-c, as outlined in Example 6. Data labels report the concentration (g/L) of xylitol.

DETAILED DESCRIPTION

[0020] Reference will now be made in detail to certain aspects of the disclosed subject matter, examples of which are illustrated in part in the accompanying drawings. While the disclosed subject matter will be described in conjunction with the enumerated claims, it will be understood that the exemplified subject matter is not intended to limit the claims to the disclosed subject matter.

[0021] In this document, the terms “a,” “an,” or “the” are used to include one or more than one unless the context clearly dictates otherwise. The term “or” is used to refer to a nonexclusive “or” unless otherwise indicated. All publications, patents, and patent documents referred to in this document are incorporated by reference herein in their entirety, as though individually incorporated by reference. In the event of inconsistent usages between this document and those documents so incorporated by reference, the usage in the incorporated reference should be considered supplement-

tary to that of this document; for irreconcilable inconsistencies, the usage in this document controls.

[0022] Values expressed in a range format should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range were explicitly recited. For example, a range of “about 0.1% to about 5%” or “about 0.1% to 5%” should be interpreted to include not just about 0.1% to about 5%, but also the individual values (e.g., 1%, 2%, 3%, and 4%) and the sub-ranges (e.g., 0.1% to 0.5%, 1.1% to 2.2%, 3.3% to 4.4%) within the indicated range. The statement “about X to Y” has the same meaning as “about X to about Y,” unless indicated otherwise. Likewise, the statement “about X, Y, or about Z” has the same meaning as “about X, about Y, or about Z,” unless indicated otherwise.

[0023] Unless expressly stated, ppm (parts per million), percentage, and ratios are on a by weight basis. Percentage on a by weight basis is also referred to as wt % or % (wt) below.

[0024] This disclosure relates to various recombinant cells engineered to produce xylitol. In general, the recombinant cells described herein have an active pentose phosphate pathway and are characterized by expression of an exogenous sugar phosphatase (XKS) enzyme. The recombinant cells described herein may optionally also express an exogenous xylitol dehydrogenase (XDH) enzyme. The disclosure further provides fermentation methods for the production of xylitol from dextrose using the genetically engineered cells described herein.

[0025] In general, recombinant cells described herein are yeast cells. As used herein, “yeast” refers to eukaryotic single celled microorganisms classified as members of the fungus kingdom. Yeast are unicellular organisms which evolved from multicellular ancestors with some species retaining multicellular characteristics such as forming strings of connected budding cells known as pseudo hyphae or false hyphae. Yeast cells may also be referred to in the art as yeast-like cells, and as used herein “yeast cell” encompasses both yeast and yeast-like cells. Suitable yeast and yeast-like host cells for modification may include, but are not limited to, *Saccharomyces cerevisiae*, *Komagataella* sp., *Kluyveromyces* (e.g., *Kluyveromyces lactis*, *Kluyveromyces marxianus*), *Yarrowia lipolytica*, *Issatchenkia orientalis*, *Pichia galeiformis*, *Pichia* sp. YB-4149 (NRRL designation), *Pichia pastoris*, *Candida* (e.g., *Candida magnoliae*, *Candida ethanolica*), *Pichia deserticola*, *Pichia membrani-fadens*, *Pichia fermentans*, *Aspergillus*, *Trichoderma*, *Myceliphthora thermophila*, *Moniliella* (e.g., *Moniliella pollinis*), *Pfaffia*, *Yamadazyma*, *Hansenula*, *Pichia kudriavzevii*, *Trichosporonoides* (e.g., *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*), *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Penicillium*, and *Torula*. An ordinarily skilled artisan would understand the requirements for selection of a suitable yeast cell, and recombinant yeast cells of the present disclosure are not limited to those expressly recited herein. Methods for genetic engineering of yeast cells are known and described in the art and a skilled artisan would understand the methods necessary to transform and engineer a suitable yeast cell.

[0026] A suitable yeast cell may be a cell of the phylum Basidiomycota and the subphylum Ustilaginomycotina. Suitable yeast of the subphylum Ustilaginomycotina

include, but are not limited to, *Ustilago* (e.g., *U. cynodontis*, *U. maydis*, *U. sphaerogena*, *U. cordal*, *U. scitaminea*, *U. coicis*, *U. syntherismae*, *U. esculenta*, *U. neglecta*, *U. crus-galli*, *Ustilago avenae*), *Sporisorium* (e.g., *Sporisorium exsertum*), *Moniliella* (e.g., *M. pollinis*, *M. tomentosa*, *M. acetoabutans*, *M. fonsecae*, *M. madida*, *M. megachiliensis*, *M. oedocephalis*, *M. nigrescens*), and *Pseudozyma* (e.g., *Pseudozyma tsukubaensis*), and *Trichosporonoides* (e.g., *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*). Yeast of the subphylum Ustilaginomycotina have been known and described in the art as potential production organisms for valuable chemicals such as itaconate, malate, succinate, mannitol, and erythritol and other valuable biotechnological applications. See, for example, Geiser et al. (Prospecting the biodiversity of the fungal family Ustilaginaceae for the production of value-added chemicals,” *Fungal Biol Biotechnol*, 2014, 1:2), Feldbrugge et al., (“The biotechnological use and potential of plant pathogenic smut fungi,” *Appl Microbiol Biotechnol*, 2013, 97(8):3253-65), Guevarra et al., (“Accumulation of itaconic, 2-hydroxyparaconic, itatartaric, and malic acids by strains of the genus *Ustilago*, *Agric. Biol. Chem.*, 1990, 54(9), 2353-2358), and Moon et al., (“Biotechnological production of erythritol and its applications,” *Appl Microbiol Biotechnol*, 2010, 86:1017-1025).

[0027] A suitable yeast cell will have an active pentose phosphate pathway that produces ribulose-5-phosphate. As used herein “active pentose phosphate pathway” refers to expression of one or more functional enzymes which, together, convert glucose-6-phosphate, NADP⁺ or NAD⁺, and water to NADPH or NADH, CO₂, and ribulose-5-phosphate. Continuing in a non-oxidative phase, the pathway may also produce other pentose (i.e., 5-carbon) sugars. For example, the pentose phosphate pathway may produce ribulose-5-phosphate, ribose-5-phosphate, xylulose-5-phosphate, fructose 6-phosphate, combinations thereof, and the like, depending on the enzymatic activities present. The active pentose phosphate pathway may be native to the yeast cell, or it may be introduced into the yeast cell by genetic engineering.

[0028] The yeast cell may be an osmotolerant yeast cell. As used herein, “osmotolerant” refers to a yeast capable of growth and reproduction under conditions of high osmolarity, such as at least 10% (w/v), at least 20% (w/v), at least 30% (w/v), at least 40% (w/v), at least 50% (w/v), or at least 60% (w/v) glucose and/or at least 6% (w/v), at least 10% (w/v), at least 12% (w/v), at least 13% (w/v), at least 15% (w/v) sodium chloride. Species and strains of osmotolerant yeast are known and described in the art, including many species of yeast used in industrial fermentation processes. Likewise, methods for assaying yeast osmotolerance are known and described in the art. See, for example, Tiwari, S., et al., (“Nectar yeast community of tropical flowering plants and assessment of their osmotolerance and xylitol-producing potential,” *Current Microbiology*, 2022, 79:28).

[0029] The recombinant yeast cell may be a recombinant *Moniliella* cell, for example, a *Moniliella pollinis* cell. FIG. 1 shows the predicted native pentose phosphate and glycolysis pathways in *Moniliella pollinis*. *Moniliella* has previously been used in the fermentation production of erythritol and methods for genetically modifying and fermenting *Moniliella* are known and described in the art. See, for example, Li et al. (“Methods for genetic transformation of filamentous fungi,” 2017, *Microb Cell Fact*, 16:168).

[0030] Various plasmids and methods for transformation of *Moniliella* are also described in the Examples below. For example, *Moniliella* may be transformed using a bipartite polynucleotide sequence in which, following recombination, the exogenous polynucleotide of interest is integrated at the specified locus and the selection marker is expressible within the cell. Suitable selection markers are known and used in the art. The selectable marker may include, but is not limited to, amdS (for example broken into a 3' portion, SEQ ID NO:91, and a 5' portion, SEQ ID NO:92), G418 resistance gene (for example broken into a 3' portion, SEQ ID NO:97, and a 5' portion, SEQ ID NO:98), zeocin resistance gene (for example broken into a 3' portion, SEQ ID NO:93, and a 5' portion, SEQ ID NO:94), nourseothricin N-acetyl transferase (NAT) (for example broken into a 3' portion, SEQ ID NO:95, and a 5' portion, SEQ ID NO:96), and invertase gene (SUC2) (for example a 3' portion of SEQ ID NO:99 and a 5' portion of SEQ ID NO:100).

[0031] The recombinant cells described herein include one or more exogenous polynucleotide sequences encoding one or more exogenous polypeptides that, when expressed improve the fermentation of glucose to ribitol by the recombinant cells.

[0032] The terms “glucose” and “dextrose” are used interchangeably herein and refer to D-glucose except where expressly indicated otherwise.

[0033] As used herein, “exogenous” refers to genetic material or an expression product thereof that originates from outside of the host organism. For example, the exogenous genetic material or expression product thereof can be a modified form of genetic material native to the host organism, it can be derived from another organism, it can be a modified form of a component derived from another organism, or it can be a synthetically derived component. For example, a *K. lactis* invertase gene is exogenous when introduced into *S. cerevisiae*.

[0034] As used herein, “native” refers to genetic material or an expression product thereof that is found, apart from individual-to-individual mutations which do not affect function or expression, within the genome of wild-type cells of the host cell. For the purposes of this application, the *Moniliella pollinis* cell “*Moniliella tomentosa* var *pollinis* TCV364” described in U.S. Pat. No. 6,440,712, which is incorporated herein by reference in its entirety, and deposited under the Budapest Treaty at BCCM/MUCL (Belgian Coordinated Collections of Micro-organisms/Mycothèque de l'Université Catholique de Louvain by *Eridania* Béghin Say, Vilvoorde R&D Centre, Havenstraat 84, B-1800 Vilvoorde) on Mar. 28, 1997 under number MUCL40385, is considered the wild-type *Moniliella pollinis* cell.

[0035] As used herein, the terms “polypeptide” and “peptide” are used interchangeably and refer to the collective primary, secondary, tertiary, and quaternary amino acid sequences and structure necessary to give the recited macromolecule its function and properties. As used herein, “enzyme” or “biosynthetic pathway enzyme” refer to a protein that catalyzes a chemical reaction. The recitation of any particular enzyme, either independently or as part of a biosynthetic pathway is understood to include the co-factors, co-enzymes, and metals necessary for the enzyme to properly function. A summary of the amino acids and their three and one letter symbols as understood in the art is presented in Table 1. The amino acid name, three letter symbol, and one letter symbol are used interchangeably herein.

TABLE 1

Amino Acid three and one letter symbols		
Amino Acid	Three letter symbol	One letter symbol
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

[0036] Variants or sequences having substantial identity or homology with the polypeptides described herein can be utilized in the practice of the disclosed recombinant cells, compositions, and methods. Such sequences can be referred to as variants or modified sequences. That is, a polypeptide sequence can be modified yet still retain the ability to exhibit the desired activity. Generally, the variant or modified sequence may include greater than about 45%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% sequence identity with the wild type, naturally occurring polypeptide sequence, or with a variant polypeptide as described herein.

[0037] As used herein, the phrases “% sequence identity,” “% identity,” and “percent identity,” are used interchangeably and refer to the percentage of residue matches between at least two amino acid sequences or at least two nucleic acid sequences aligned using a standardized algorithm. Methods of amino acid and nucleic acid sequence alignment are well-known. Sequence alignment and generation of sequence identity include global alignments and local alignments which are carried out using computational approaches. An alignment can be performed using BLAST (National Center for Biological Information (NCBI) Basic Local Alignment Search Tool) version 2.2.31 software with default parameters. Amino acid % sequence identity between amino acid sequences can be determined using standard protein BLAST with the following default parameters: Max target sequences: 100; Short queries: Automatically adjust parameters for short input sequences; Expect threshold: 10; Word size: 6; Max matches in a query range: 0; Matrix: BLOSUM62; Gap Costs: (Existence: 11, Extension: 1); Compositional adjustments: Conditional compositional score matrix adjustment; Filter: none selected; Mask: none selected. Nucleic acid % sequence identity between nucleic acid sequences can be determined using standard nucleotide BLAST with the following default parameters: Max target sequences: 100; Short queries: Automatically adjust parameters for short input sequences; Expect threshold: 10; Word size: 28; Max matches in a query range: 0; Match/Mismatch Scores: 1, -2; Gap costs: Linear; Filter: Low complexity regions; Mask: Mask for lookup table only.

A sequence having an identity score of XX % (for example, 80%) with regard to a reference sequence using the NCBI BLAST version 2.2.31 algorithm with default parameters is considered to be at least XX % identical or, equivalently, have XX % sequence identity to the reference sequence.

[0038] Polypeptide or polynucleotide sequence identity may be measured over the length of an entire defined polypeptide sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined polypeptide sequence, for instance, a fragment of at least 15, at least 20, at least 30, at least 40, at least 50, at least 70 or at least 150 contiguous residues. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

[0039] The polypeptides disclosed herein may include “variant” polypeptides, “mutants,” and “derivatives thereof.” As used herein the term “wild-type” is a term of the art understood by skilled persons and means the typical form of a polypeptide as it occurs in nature as distinguished from variant or mutant forms. As used herein, a “variant,” “mutant,” or “derivative” refers to a polypeptide molecule having an amino acid sequence that differs from a reference protein or polypeptide molecule. A variant or mutant may have one or more insertions, deletions, or substitutions of an amino acid residue relative to a reference molecule.

[0040] The amino acid sequences of the polypeptide variants, mutants, derivatives, or fragments as contemplated herein may include conservative amino acid substitutions relative to a reference amino acid sequence. For example, a variant, mutant, derivative, or fragment polypeptide may include conservative amino acid substitutions relative to a reference molecule. “Conservative amino acid substitutions” are those substitutions that are a substitution of an amino acid for a different amino acid where the substitution is predicted to interfere least with the properties of the reference polypeptide. In other words, conservative amino acid substitutions substantially conserve the structure and the function of the reference polypeptide. Conservative amino acid substitutions generally maintain (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a beta sheet or alpha helical conformation, (b) the charge and/or hydrophobicity of the molecule at the site of the substitution, and/or (c) the bulk of the side chain.

[0041] As used herein, terms “polynucleotide,” “polynucleotide sequence,” and “nucleic acid sequence,” and “nucleic acid,” are used interchangeably and refer to a sequence of nucleotides or any fragment thereof. These phrases also refer to DNA or RNA of natural or synthetic origin, which may be single-stranded or double-stranded and may represent the sense or the antisense strand. The DNA polynucleotides may be a cDNA (e.g., coding DNA) or a genomic DNA sequence (e.g., including both introns and exons).

[0042] A polynucleotide is said to encode a polypeptide if, in its native state or when manipulated by methods known to those skilled in the art, it can be transcribed and/or translated to produce the polypeptide or a fragment thereof. The anti-sense strand of such a polynucleotide is also said to encode the sequence.

[0043] Those of skill in the art understand the degeneracy of the genetic code and that a variety of polynucleotides can encode the same polypeptide. In some aspects, the polynucleotides (e.g., polynucleotides encoding an XKS or XDH polypeptide) may be codon-optimized for expression in a particular cell including, without limitation, a plant cell, bacterial cell, fungal cell, or animal cell. While polypeptides encoded by polynucleotide sequences found in various species are disclosed herein any polynucleotide sequences may be used which encodes a desired form of the polypeptides described herein. Thus, non-naturally occurring sequences may be used. These may be desirable, for example, to enhance expression in heterologous expression systems of polypeptides or proteins. Computer programs for generating degenerate coding sequences are available and can be used for this purpose. Pencil, paper, the genetic code, and a human hand can also be used to generate degenerate coding sequences.

[0044] The recombinant cells described herein may include deletions or disruptions in one or more native genes. The phrase “deletion or disruption” refers to the status of a native gene in the recombinant cell that has either a completely eliminated coding region (deletion) or a modification of the gene, its promoter, or its terminator (such as by a deletion, insertion, or mutation) so that the gene no longer produces an active expression product, produces severely reduced quantities of the expression product (e.g., at least a 75% reduction or at least a 90% reduction) or produces an expression product with severely reduced activity (e.g., at least 75% reduced or at least 90% reduced). The deletion or disruption can be achieved by genetic engineering methods, forced evolution, mutagenesis, RNA interference (RNAi), and/or selection and screening. The native gene to be deleted or disrupted may be replaced with an exogenous nucleic acid of interest for the expression of an exogenous gene product (e.g., polypeptide, enzyme, and the like).

[0045] The recombinant cells described herein may include one or more genetic modifications in which an exogenous nucleic acid is integrated into the genome of the host cell. One of skill in the art know how to select suitable loci in a yeast genome for integration of the exogenous nucleic acid. Suitable integration loci may include, but are not limited to, the PDC1, GPD1, CYB2A, CYB2B, g4240, YMR226, MDHB, ATO2, Adh9091, Adh1202, ADE2, ADH2556, GAL6, MDH1, SCW11, ER1, ER3, pyrF, TRP3, gpdIIA, and gpdIIB loci. For example, in a *M. pollinis* host cells, suitable interaction loci may include, but are not limited to, the ER1 locus (defined as the locus flanked by SEQ ID NO:87 and SEQ ID NO:88), the ER3 locus (defined as the locus flanked by SEQ ID NO:81 and SEQ ID NO:82), the PDC1 locus (defined as the locus flanked by SEQ ID NO:75 and SEQ ID NO:76), the pyrF locus (defined as the locus flanked by SEQ ID NO:77 and SEQ ID NO:78), the TRP3 locus (defined as the locus flanked by SEQ ID NO:83 and SEQ ID NO:84), the gpdIIA locus (defined as the locus flanked by SEQ ID NO:85 and SEQ ID NO:86); and the gpdIIB locus (defined as the locus flanked by SEQ ID NO:89 and SEQ ID NO:90). The exogenous nucleic acid may also be integrated in an intergenic region or other location in the host cell genome not specifically specified herein. Other suitable integration loci may be determined by one of skill in the art. Furthermore, one of skill in the art would recognize how to use sequences to design primers to verify correct gene integration at the chosen locus.

[0046] The recombinant cell may have one or more copies of a given exogenous nucleic acid sequence integrated in a host chromosome(s) and replicated together with the chromosome(s) into which it has been integrated. For example, the yeast cell may be transformed with nucleic acid construct including a polynucleotide sequence encoding for a polypeptide described herein and the polynucleotide sequence encoding for the polypeptide may be integrated in one or more copies in a host chromosome(s). The recombinant cell may include multiple copies (two or more) of a given polynucleotide sequence encoding a polypeptide described herein. The recombinant cell may have one, two, three, four, five, six, seven, eight, nine, ten, or more copies of a polynucleotide sequence encoding a polypeptide described herein integrated into the genome. The multiple copies of said polynucleotide sequence may all be incorporated at a single locus or may be incorporated at multiple loci.

[0047] The recombinant cells described herein are capable of producing xylitol and include an exogenous polynucleotide sequence encoding a xylulokinase (XKS) enzyme. The exogenous polynucleotide sequence may be an exogenous xylulose sugar phosphatase (XKS) gene.

[0048] A “xylulokinase gene” and an “XKS gene” are used interchangeably herein and refer to any gene or polynucleotide that encodes a polypeptide with xylulokinase activity. As used herein, “xylulokinase activity” refer to the ability to catalyze the conversion of xylulose-5-phosphate and ADP to xylulose and ATP. The XKS gene may be derived from any suitable source. For example, the XKS gene may be derived from *Saccharomyces cerevisiae*. The XKS gene may encode an amino acid at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, or 100% sequence identity to the amino acid sequence of at least one of SEQ ID NOs:8 and 20.

[0049] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Saccharomyces cerevisiae* DOG1 sugar phosphatase gene encoding the amino acid of SEQ ID NO:8. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:8.

[0050] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Saccharomyces cerevisiae* DOG2 sugar phosphatase gene encoding the amino acid of SEQ ID NO:20. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:20.

[0051] Recombinant cells described herein may additionally include an exogenous polynucleotide sequence encoding a xylitol dehydrogenase (XDH) enzyme. The exogenous polynucleotide sequence may be an exogenous xylitol dehydrogenase (XDH) gene.

[0052] A “xylitol dehydrogenase gene” and an “XDH gene” are used interchangeably herein and refer to any gene or polynucleotide that encodes a polypeptide with xylitol dehydrogenase activity. As used herein, “xylitol dehydrogenase activity” refer to the ability to catalyze the conversion of xylulose and NADH or NADPH to xylitol and NAD⁺ or

NADP⁺. The XDH gene may be derived from any suitable source. For example, the XDH gene may be derived from *Pichia stipitis*, *Rhodobacteraceae bacterium*, or *Bemisia argentifolii*. The XDH gene may encode an amino acid at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, or 100% sequence identity to the amino acid sequence of at least one of SEQ ID NOs:1, 13, or 14.

[0053] The recombinant cell may comprise an exogenous polynucleotide that is, or may be derived from, a cofactor switched *Pichia stipitis* XDH gene encoding the amino acid of SEQ ID NO:1. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:1.

[0054] The recombinant cell may comprise an exogenous polynucleotide that is, or may be derived from, a *Rhodobacteraceae bacterium* SDR family oxidoreductase gene encoding the amino acid of SEQ ID NO:13. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:13.

[0055] The recombinant cell may comprise an exogenous polynucleotide that is, or may be derived from, a *Bemisia argentifolii* (Silverleaf Whitefly) ketose reductase (sorbitol dehydrogenase) gene encoding the amino acid of SEQ ID NO:14. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:14.

[0056] The exogenous polynucleotides in the recombinant cells described herein may be under the control of a promoter. For example, the exogenous nucleic acid may be operably linked to a heterologous or artificial promoter. Suitable promoters are known and described in the art. Promoters may include, but are not limited to, pyruvate decarboxylase promoter (PDC), translation elongation factor 2 promoter (TEF2), SED1, alcohol dehydrogenase 1A promoter (ADH1), hexokinase 2 promoter (HXK2), FLO5 promoter, pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:61); 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:52); glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:54); translational elongation factor 1 promoter (TEFp; SEQ ID NO:55); modified TEFp (SEQ ID NO:53); phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:56); 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:57); enolase promoter (ENO1p; SEQ ID NO:58); asparagine synthetase promoter (ASNSp; SEQ ID NO:59); 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:61); and RPL16B (SEQ ID NO:62).

[0057] The exogenous nucleic acids in the recombinant cells described herein may be under the control of a terminator. For example, the exogenous nucleic acid may be operably linked to a heterologous or artificial terminator. Suitable terminators are known and described in the art. Terminators may include, but are not limited to, GAL10 terminator, PDC terminator, transaldolase terminator (TAL), 6PGD terminator (6PGDt; SEQ ID NO:63); ASNS terminator (ASNSt; SEQ ID NO:64); ENO1 terminator (ENO1t;

SEQ ID NO:65); hexokinase 1 terminator (HXK1t; SEQ ID NO:66); PGK1 terminator (PGK1t; SEQ ID NO:67); PGM1 terminator (PGM1t; SEQ ID NO:68); PYK1 terminator (PYK1t; SEQ ID NO:69); RPLA terminator (RPLAt; SEQ ID NO:70); transaldolase 1 terminator (TAL1t; SEQ ID NO:71); TDH3 terminator (TDH3t; SEQ ID NO:72); translation elongation factor 2 terminator (TEF2t; SEQ ID NO:73); triosephosphate isomerase 1 terminator (TPI1t; SEQ ID NO:74); and MpTEF1t (SEQ ID NO:101).

[0058] A promoter or terminator is “operably linked” to a given polynucleotide (e.g., a gene) if its position in the genome or expression cassette relative to said polynucleotide is such that the promoter or terminator, as the case may be, performs its transcriptional control function.

[0059] The polypeptides described herein may be provided as part of a construct. As used herein, the term “construct” refers to recombinant polynucleotides including, without limitation, DNA and RNA, which may be single-stranded or double-stranded and may represent the sense or the antisense strand. Recombinant polynucleotides are polynucleotides formed by laboratory methods that include polynucleotide sequences derived from at least two different natural sources or they may be synthetic. Constructs thus may include new modifications to endogenous genes introduced by, for example, genome editing technologies. Constructs may also include recombinant polynucleotides created using, for example, recombinant DNA methodologies. The construct may be a vector including a promoter operably linked to the polynucleotide encoding a polypeptide as described herein. As used herein, the term “vector” refers to a polynucleotide capable of transporting another polynucleotide to which it has been linked. The vector may be a plasmid, which refers to a circular double-stranded DNA loop into which additional DNA segments may be integrated.

[0060] The disclosure also provides fermentation methods for the production of xylitol using the recombinant cells described herein. The fermentation methods include the step of fermenting a substrate using the genetically engineered yeasts described herein to product xylitol. The fermentation method can include additional steps, as would be understood by a person skilled in the art. Non-limiting examples of additional process steps include maintaining the temperature of the fermentation broth within a predetermined range, adjusting the pH during fermentation, and isolating the xylitol from the fermentation broth. The fermentation process may be a fully aerobic process or a partially aerobic process.

[0061] The fermentation method can be run using a suitable fermentation substrate. The substrate of the fermentation method can include glucose, sucrose, galactose, mannose, molasses, xylose, fructose, hydrolysates of starch, lignocellulosic hydrolysates, or a combination thereof. One skilled in the art will recognize what fermentation substrate is suitable for a given fermentation organism and system.

[0062] The fermentation process can be run under various conditions. The fermentation temperature, i.e., the temperature of the fermentation broth during processing, may be ambient temperature. Alternatively, or additionally, the fermentation temperature may be maintained within a predetermined range. For example, the fermentation temperature can be maintained in the range of 25° C. to 45° C., 30° C. to 40° C., or 32° C. to 37° C., preferably about 35° C. However, a skilled artisan will recognize that the fermenta-

tion temperature is not limited to any specific range or temperature recited herein and may be modified as appropriate.

[0063] The fermentation process can be run within certain oxygen uptake rate (OUR) ranges. The volumetric OUR of the fermentation process can be in the range of 0.5 to 40, 1 to 35, 2 to 30, 3 to 25, 4 to 20, or 5 to 15 mmol O₂/(L·h). In some embodiments, the specific OUR can be in the range of 0.05 to 10, 0.1 to 8, 0.15 to 5, 0.2 to 1, or 0.3 to 0.75 mmol O₂/(g cell dry weight·h). However, the volumetric or specific OURs of the fermentation process are not limited to any specific rates or ranges recited herein.

[0064] The fermentation process can be run at various cell concentrations. In some embodiments, the cell dry weight at the end of fermentation can be 5 to 40, 8 to 30, or 10 to 20 g cell dry weight/L. Further, the pitch density or pitching rate of the fermentation process can vary. In some embodiments, the pitch density can be 0.05 to 11, 0.1 to 10, or 0.25 to 8 g cell dry weight/L.

[0065] The initial dextrose concentration of the fermentation may be at least 100, 200, 250, 300, 350, or at least 400 g/L dextrose. The initial dextrose concentration may be between 100 to 400, 150 to 350, or 250 to 325 g/L.

[0066] The fermentation process can be associated with various characteristics, such as, but not limited to, fermentation production rate, pathway fermentation yield, final titer, and peak fermentation rate. These characteristics can be affected by the selection of the yeast and/or genetic modification of the yeast used in the fermentation process. These characteristics can be affected by adjusting the fermentation process conditions. These characteristics can be adjusted via a combination of yeast selection or modification and the selection of fermentation process conditions.

[0067] The xylitol production rate of the process may be at least 0.2, 0.3, 0.5, 0.75, or at least 1.0 g L⁻¹ h⁻¹. The final xylitol titer of the process may be at least 0.5, 1.0, 1.5, 2.0, 2.5 or 3.0 g/L.

[0068] The fermentation process can be run as a dextrose-fed batch. Further, the fermentation process can be a batch process, continuous process, or semi-continuous process, as would be understood by a person skilled in the art.

EXAMPLES

[0069] The invention is further described in detail by reference to the following experimental examples. These examples are provided for purposes of illustration only and are not intended to be limiting unless otherwise specified. Thus, the invention should in no way be construed as being limited to the following examples, but rather should be construed to encompass any and all variations which become evident as a result of the teaching provided herein.

Example 1—Genetically Modified *Moniliella pollinis* Strains

[0070] Strain 1-1 is the *Moniliella pollinis* host strain “*Moniliella tomentosa* var *pollinis* TCV364” described in U.S. Pat. No. 6,440,712, which is incorporated herein by reference in its entirety, and deposited under the Budapest Treaty at BCCM/MUCL (Belgian Coordinated Collections of Micro-organisms/Mycothèque de l’Université Catholique de Louvain by *Eridania* Béghin Say, Vilvoorde R&D Centre, Havenstraat 84, B-1800 Vilvoorde) on Mar. 28, 1997, under number MUCL40385. Table 2 below lists various

Moniliella pollinis strains, including information on the parent strain, the sequence with which the parent strain was transformed, and characterizations of the expression cassette (s) contained on the transformed sequence.

[0071] In Table 2, the following cassettes were utilized. Each “XHD Homolog+3’ G418 Expression Cassette” contained, in order, a 5’ ER1 flanking sequence (SEQ ID NO:87), a MpPGM1 promoter (SEQ ID NO:56), a gene encoding the indicated XDH enzyme homolog (one of SEQ ID NOs:1, 13, or 14), a MpPGK1 terminator (SEQ ID NO:67), and a 3’ portion of a G418 resistance gene (SEQ ID NO:97). Each “XKS/Phosphatase Homolog+5’ G418 Expression Cassette” contained, in order, a 5’ portion of a G418 resistance gene (SEQ ID NO:98), a MpPGK promoter (SEQ ID NO:57), a gene encoding the indicated XKS/Phosphatase enzyme homolog (one of SEQ ID NOs:35 or 41-50), a MpENO1 terminator (SEQ ID NO:65), and a 3’ ER1 flanking sequence (SEQ ID NO:88).

[0072] In Table 3, the following cassettes were utilized. Each “XDH Homolog Expression Cassette” contained, in order, a 5’ ER1 flanking sequence (SEQ ID NO:87), a MpPYK1 promoter (SEQ ID NO:61), a gene encoding the indicated XDH homolog (one of SEQ ID NOs:15-19), a Mp6PGD terminator (SEQ ID NO:63), and a 5’ portion of a Zeocin resistance gene (SEQ ID NO:93). The “Selection Marker Cassette” of SEQ ID NO:38 contained, in order, a 3’ portion of a Zeocin selection marker gene (SEQ ID NO:93), an MpTEF2 terminator (SEQ ID NO:73), and a 3’ ER1 flanking sequence (SEQ ID NO:88). Each “XKS/Phosphatase Expression Cassette” contained, in order, a 5’ ER1 flanking sequence (SEQ ID NO:87), a PGK1 promoter (SEQ ID NO:67), a gene encoding the indicated XKS/Phosphatase enzyme homolog (one of SEQ ID NOs:8 or 20-24), MpENO1 terminator (SEQ ID NO:65), a 3’ portion of either a G418 (SEQ ID NO:97) or a Zeocin (SEQ ID NO:93) resistance gene, and an MpTEF2 terminator (SEQ ID NO:73). The “Selection Marker Cassette” of SEQ ID NO:37 contained, in order, a 5’ portion of a Zeocin resistance gene and a 3’ ER1 flanking sequence (SEQ ID NO:88). The “Selection Marker Cassette” of SEQ ID NO:36 contained, in order, a 5’ portion of a G418 resistance gene and a 3’ ER1 flanking sequence (SEQ ID NO:88).

[0073] Upon bipartite transformation with both cassettes as indicated in Tables A and B, the two cassettes recombine for integration of both the nucleotide sequence encoding the indicated XDH or XKS/phosphatase homolog and the indicated selectable marker at the ER1 loci.

[0074] The indicated *Moniliella pollinis* parent strain was transformed with the indicated sequence(s) by first protoplasting the parent strain by adding an enzyme mixture containing 0.6M MgSO₄, 7.5 g/L driselase, and 12.5 g/L *Trichoderma harzianum* lysing enzyme to a mycelial pellet of the parent strain. Protoplasts were then pelleted, washed with 0.6M MgSO₄, and resuspended in STC medium (0.6M sucrose, 50 mM CaCl₂, 10 mM Tris-HCl, pH 7.5). Bipartite transformations were prepared by adding 100 µg single stranded salmon sperm DNA and 1.5 to 5 µg each of the 5’ and 3’ DNA transformation fragments (3-10 µg total; see Table 4 for list of fragments) to approximately 200 µL protoplast mixture (10⁸ cells/mL). 1 mL 50% PEG in STC medium was then added to the salmon sperm DNA, transformation DNA, and protoplast mixture and the resulting combination was incubated for 15 minutes at room temperature. Following incubation, recovery broth (0.4M sucrose, 1

g/L yeast extract, 1 g/L malt extract, 10 g/L glucose, pH 4.5) was added to the mixture and incubated at 27° C., 100 rpm, for 16 to 24 hours. Following the incubation, protoplasts were pelleted by centrifugation and resuspended in 1 mL PBS.

[0075] The resuspended protoplasts were plated on PDA+250 mg/L geneticin (G418) or PDA+100 mg/L zeocin selection plates and incubated at 30-35° C. for at least 2-4 days until transformants grow. Resulting transformants were evaluated by colony PCR for integration of the indicated sequence. A PCR verified isolate was then designated as the indicated strain number. In some instances, more than one PCR verified isolate, e.g., “sister” isolates, are indicated by letters following the strain number. For example, strain 1-2 has 3 sister isolates, strains 1-2a, 1-2b, and 1-2c.

[0076] For example, Strain 1-1 was transformed with SEQ ID NO:39 and SEQ ID NO:41. SEQ ID NO:39 contains i)

a 5' ER1 flanking sequence (SEQ ID NO:87), ii) a MpPGM1 promoter (SEQ ID NO:56), iii) an expression cassette for the cofactor switched XDH from *Pichia stipitis* of SEQ ID NO:1, iv) a MpPGK1 terminator (SEQ ID NO:67), and v) a 3' portion of a G418 resistance gene (SEQ ID NO:97). SEQ ID NO:41 contains i) a 5' portion of a G418 resistance gene (SEQ ID NO:98), ii) a MpPGK promoter (SEQ ID NO:57), iii) an expression cassette encoding the *Hypsizygus marmoreus* XKS1 of SEQ ID NO:2, iv) a MpENO1 terminator (SEQ ID NO:65), and v) a 3' ER1 flanking sequence (SEQ ID NO:88). Transformants were selected on PDA+250 mg/L geneticin (G418) selection plates and incubated at 35° C. for at least 2 days until transformants grow. Resulting transformants were streaked for single colony isolation on PDA+geneticin (G418) plates and single colonies were selected. Selected colonies were evaluated by colony PCR for integration of the indicated sequence. PCR verified isolates were designated strains 1-2a, 1-2b, and 1-2c.

TABLE 2

Strain	Parent Strain	Transformation SEQ ID NO:	XDH Homolog + 3' G418 Expression Cassette		Encoded Polypeptide SEQ ID NO:	XKS/Phosphatase Homolog + 5' G418 Expression Cassette		Encoded Polypeptide SEQ ID NO:
			Polypeptide Source Organism (Gene Name and/or Accession Number)			Polypeptide Source Organism (Gene Name and/or Accession Number)		
1-2a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Hypsizygus marmoreus</i> XKS1		2
1-3a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Cylindrobasidium torrendii</i> XKS1		3
1-4a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Pleurotus ostreatus</i> XKS1		4
1-5a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Trametes coccinea</i> XKS1		5
1-6a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Antridiella citronella</i> XKS1		6
1-7a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Ustilago maydis</i> XKS1		7
1-8a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Saccharomyces cerevisiae</i> DOG1		8
1-9a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Bacteroides thetaiotaomicron</i> XKS1		9
1-10a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Pichia stipitis</i> XKS		10
1-11a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Saccharomyces cerevisiae</i> XKS		11
1-12a-c	1-1	39	<i>Pichia stipitis</i> cofactor switched XDH		1	<i>Streptomyces avermitilis</i> HAD		12
1-13	1-1	40	<i>Rhodobacteraceae bacterium</i> SDR family oxidoreductase		13	<i>Saccharomyces cerevisiae</i> DOG1		8
1-14a-c	1-1	51	<i>Bemisia argentifolii</i> (Silverleaf Whitefly) ketose reductase (sorbitol dehydrogenase)		14	<i>Saccharomyces cerevisiae</i> DOG1		8

TABLE 3

Strain	Parent Strain	Transformation SEQ ID NO:	XDH Homolog Expression Cassette		Transformation SEQ ID NO:	XKS/Phosphatase Expression Cassette		Transformation SEQ ID NO:
			Polypeptide Source Organism (Gene Name and/or Accession Number)	Encoded Polypeptide		Polypeptide Source Organism (Gene Name and/or Accession Number)	Encoded Polypeptide	
1-15a-d	1-14c	25	<i>Bos indicus</i> × <i>Bos taurus</i> XDH (XP_027409920)	15	—	—	—	38

TABLE 3-continued

Strain	Parent Strain	XDH Homolog Expression Cassette			XKS/Phosphatase Expression Cassette			Selectable Marker Cassette	
		Transformation SEQ ID NO:	Polypeptide Source Organism (Gene Name) and/or Accession Number	Encoded Polypeptide SEQ ID NO:	Transformation SEQ ID NO:	Polypeptide Source Organism (Gene Name) and/or Accession Number	Encoded Polypeptide SEQ ID NO:	Selectable Marker	Transformation SEQ ID NO:
1-16a-d	1-14c	26	<i>Aspergillus tubingensis</i> XDH (GFN13953)	16	—	—	—	Zeocin	38
1-17a-e	1-14c	27	<i>Chaetomium thermophilum</i> var. <i>thermophilum</i> DSM 1495 XDH (EGS17058)	17	—	—	—	Zeocin	38
1-18a-e	1-14c	28	<i>Candidatus Poribacteria bacterium</i> XDH (RKY04443)	18	—	—	—	Zeocin	38
1-19a-e	1-14c	29	<i>Linepithema humile</i> XDH (XP_012217409)	19	—	—	—	Zeocin	38
1-20a-e	1-14c	—	—	—	30	<i>Saccharomyces cerevisiae</i> (KZV10766) Sugar Phosphatase DOG2	20	Zeocin	37
1-21a-e	1-14c	—	—	—	31	<i>Acidobacteria bacterium</i> Sugar Phosphatase (PYX10474)	21	Zeocin	37
1-22	1-14c	—	—	—	32	<i>Mesorhizobium</i> sp. Sugar Phosphatase (TIO13298)	22	Zeocin	37
1-23a-e	1-14c	—	—	—	33	<i>Escherichia coli</i> Sugar Phosphatase (EFE9125297)	23	Zeocin	37
1-24a-e	1-14c	—	—	—	34	<i>Puerhobacterium puerhi</i> Sugar Phosphatase (WP_159795540)	24	Zeocin	37
1-25a-d	1-1	—	—	—	35	<i>Saccharomyces cerevisiae</i> DOG1	8	G418	36
1-26a-e	1-1	—	—	—	30	<i>Saccharomyces cerevisiae</i> (KZV10766) Sugar Phosphatase DOG2	20	Zeocin	37
1-27a-e	1-1	—	—	—	31	<i>Acidobacteria bacterium</i> Sugar Phosphatase (PYX10474)	21	Zeocin	37
1-28a-c	1-1	—	—	—	34	<i>Puerhobacterium puerhi</i> Sugar Phosphatase (WP_159795540)	24	Zeocin	37

Example 2—Shake Flask Fermentation Assay

[0077] Strains 1-2a-c, 1-3a-c, 1-4a-c, 1-5a-c, 1-6a-c, 1-7a-c, 1-8a-c, 1-9a-c, 1-10a-c, 1-11a-c, and 1-12a-c (outlined in Table 2 above), were run in shake flasks to assess xylitol, erythritol, ribitol, glycerol, arabitol, and ethanol production and glucose consumption.

[0078] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10

g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectropho-

tometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0079] A 250 ml non-baffled flask containing production medium (Table C) was inoculated with the seed culture to form the production culture with a starting OD600 of about 0.4 (approximately 0.8 mL of the seed culture). The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, arabitol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 5 and FIG. 2. The expression of both the *Pichia stipitis* cofactor switched XDH (SEQ ID NO:1) and the *Saccharomyces cerevisiae* DOG1 sugar phosphatase (SEQ ID NO:8) in *Moniliella pollinis* resulted in production of xylitol.

TABLE 4

Production Medium	
Component	Concentration (units)
Glucose	300 (g/L)
KH ₂ PO ₄	1.27 (g/L)

TABLE 4-continued

Production Medium	
Component	Concentration (units)
(NH ₄) ₂ HPO ₄	0.13 (g/L)
(NH ₄) ₂ SO ₄	1.80 (g/L)
Urea	2.85 (g/L)
Citric Acid	200.0 (mg/L)
MgSO ₄ ·7H ₂ O	690.2 (mg/L)
FeSO ₄ ·7H ₂ O	32.35 (mg/L)
ZnSO ₄ ·7H ₂ O	4.40 (mg/L)
MnSO ₄ ·H ₂ O	0.53 (mg/L)
CuSO ₄ ·5H ₂ O	0.45 (mg/L)
CaCl ₂ ·2H ₂ O	91.60 (mg/L)
Na ₂ SO ₄	154.50 (mg/L)
Thiamine-HCl	10 (mg/L)
Choline-Cl	100 (mg/L)
Antifoam CF-32	1.00 (g/L)

TABLE 5

96 hour Shake Flask Results							
Fermentation Broth Analyte (g/L)							
Strain	Glucose	Ribitol	Erythritol	Glycerol	Ethanol	Xylitol	Arabitol
1-1	177.3	0.3	21.3	2.6	24.1	0.000	0.00
1-1	221.2	0.6	16.7	8.4	9.5	0.000	0.00
1-1	251.5	0.4	9.4	6.8	3.0	0.000	0.00
1-2a	210.2	0.8	20.3	8.9	10.2	0.000	0.00
1-2b	192.1	0.6	28.0	8.4	14.6	0.000	0.00
1-2c	251.3	0.2	4.2	4.0	1.1	0.000	0.00
1-3a	214.3	1.6	13.8	9.3	12.1	0.000	0.00
1-3b	151.4	0.6	36.8	9.6	24.3	0.000	0.00
1-3c	77.0	1.6	72.9	13.4	29.9	0.000	0.00
1-4a	228.4	0.5	14.9	7.3	8.3	0.000	0.00
1-4b	283.5	0.3	3.1	2.8	1.2	0.000	0.00
1-4c	102.6	1.3	59.1	12.3	30.9	0.000	0.00
1-5a	229.0	0.5	14.9	6.7	8.3	0.000	0.00
1-5b	270.5	0.3	5.9	5.0	1.9	0.000	0.00
1-5c	278.4	0.6	4.2	3.9	1.0	0.000	0.00
1-6a	231.3	0.4	14.9	6.0	8.4	0.000	0.00
1-6b	250.1	0.4	10.2	6.4	4.0	0.000	0.00
1-6c	139.7	0.9	49.2	10.0	22.3	0.000	0.00
1-7a	287.9	0.1	2.3	1.2	2.0	0.000	0.00
1-7b	191.2	0.9	23.9	7.1	16.9	0.000	0.00
1-7c	131.5	0.9	55.7	7.6	23.5	0.000	0.00
1-8a	85.0	3.4	21.2	6.2	14.9	0.931	0.00
1-8b	105.8	7.6	52.0	10.3	27.6	2.699	0.00
1-8c	208.3	2.7	22.2	6.2	11.0	1.092	0.00
1-9a	187.7	0.9	25.0	8.5	17.9	0.000	0.00
1-9b	102.4	1.5	56.3	19.5	26.1	0.000	0.00
1-9c	221.1	0.8	16.6	5.7	10.8	0.000	0.00
1-10a	221.0	0.2	4.1	3.0	1.9	0.000	0.00
1-10b	309.7	0.8	0.3	0.7	0.5	0.000	0.00
1-10c	166.4	0.0	39.6	8.4	18.7	0.000	0.00
1-11a	121.9	1.2	57.4	6.2	25.2	0.000	0.00
1-11b	161.8	1.0	36.2	7.8	16.8	0.000	0.00
1-11c	126.9	1.1	49.1	13.4	26.1	0.000	0.00
1-12a	143.4	1.0	46.8	9.3	22.7	0.000	0.00
1-12b	219.8	0.7	13.7	8.9	9.4	0.000	0.00
1-12c	288.3	0.1	1.8	0.9	4.3	0.000	0.11

Example 3—Shake Flask Fermentation Assay

[0080] Strains 1-8a-c, 1-13, 1-14a and 1-14c (outlined in Table 2 above), were run in duplicate in shake flasks to assess xylitol, erythritol, ribitol, glycerol, and ethanol production and glucose consumption.

[0081] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0082] Either a 250 ml non-baffled flask or a 250 ml baffled flask containing production medium (Table B) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 48, 72, and 96 hours of incubation. Samples were analyzed for glucose, arabitol, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 6 and FIG. 2. As demon-

strated in the results, expression of the *Saccharomyces cerevisiae* DOG1 sugar phosphatase (SEQ ID NO:9) in combination with either the *Pichia stipitis* cofactor switched XDH (SEQ ID NO:1), the *Rhodobacteraceae bacterium* SDR family oxidoreductase (SEQ ID NO:13), or the *Bemisia argentifolii* (Silverleaf Whitefly) ketose reductase (sorbitol dehydrogenase) (SEQ ID NO:14) in *Moniliella pollinis* produced xylitol after as little as 48 hours of fermentation.

[0083] While all strains tested produced xylitol, the level of production can be changed by altering the aeration of the fermentation. For example, as shown in FIG. 3, with high aeration (i.e., fermentation done in a baffled flask) strain 1-14c produced about 0.5 g/L xylitol after 96 hours, however fermentation at lower aeration (i.e., in a non-baffled flask) resulted in xylitol production over 2.5 g/L.

[0084] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the sequences was not correctly targeted to the ER1 loci or multiple copies were integrated. Further analysis indicated that strains 1-14b and 1-13 include the transformed polynucleotide sequence, but it is not targeted to the ER1 locus. Strain 1-14b was not run in this shake flask experiment due to resource limitations. Further analysis also indicated that strain 1-14c included 8 copies of the polynucleotide encoding SEQ ID NO:14 and 10 copies of the polynucleotide encoding SEQ ID NO:8, but none of the sequences were correctly targeted to the ER1 locus.

TABLE 6

Shake Flask Results								
Time Baffled			Fermentation Broth Analyte (g/L)					
Strain	Point	Flask	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-8a	48	no	217.948	1.0374	0.2464	15.666	10.4687	11.1309
1-8a	48	no	218.631	1.0375	0.2338	15.3798	10.3807	11.0564
1-8b	48	no	219.0666	0.8671	0.0996	14.5724	10.0748	13.259
1-8b	48	no	213.3015	0.8602	0.1576	15.2142	10.2649	13.6549
1-8c	48	no	256.7603	0.5482	0.1941	6.6763	3.9799	5.2907
1-8c	48	no	256.422	0.5651	0.1815	6.6742	3.8904	5.3232
1-13	48	no	246.6843	0.3063	0.0831	7.9747	5.5501	7.7021
1-13	48	no	251.1016	0.3189	0.1035	8.5806	5.9284	8.0667
1-14a	48	no	227.6182	0.3365	0.047	14.9907	13.0872	1.4197
1-14a	48	no	230.0448	0.2339	0.0757	10.1985	5.5809	12.2101
1-14c	48	no	262.5613	0.5171	0.2866	6.8507	4.5567	5.6778
1-14c	48	no	257.487	0.524	0.2912	6.6819	4.047	5.7798
1-8a	48	yes	225.6687	0.4823	0.0394	17.2254	18.4131	1.2625
1-8a	48	yes	221.4495	0.501	0.0209	17.7996	18.7213	1.2621
1-8b	48	yes	201.8136	1.2746	0.0416	20.798	24.7817	2.0304
1-8b	48	yes	193.9384	1.4128	0.0485	22.8222	27.5048	1.9007
1-8c	48	yes	262.1726	0.5381	0.1465	7.9166	6.2902	1.1757
1-8c	48	yes	264.5841	0.4884	0.139	7.2657	5.4647	1.1227
1-13	48	yes	241.0328	0.3489	0.039	12.3663	11.3904	0.53
1-13	48	yes	237.0301	0.3694	0.0285	13.3439	12.7668	0.5016
1-14a	48	yes	229.0956	0.3295	0.0531	14.5186	12.3315	2.179
1-14a	48	yes	219.8367	0.3411	0.0338	16.4859	14.9275	1.7324
1-14c	48	yes	260.7984	0.4379	0.2001	7.216	5.4117	1.0756
1-14c	48	yes	260.4357	0.4595	0.2134	7.4861	5.8055	1.1902
1-8a	72	no	118.9917	5.1176	1.0347	48.021	14.6104	22.6909
1-8a	72	no	121.258	5.0107	1.0405	47.5303	14.4495	22.7773
1-8b	72	no	100.3422	2.7894	0.5144	42.2335	14.1556	34.3714
1-8b	72	no	95.5028	2.8572	0.4867	43.7333	14.2579	34.2852
1-8c	72	no	187.1468	3.0444	0.7336	27.5958	5.8653	14.635
1-8c	72	no	187.8452	3.0079	0.7989	27.3219	5.7335	14.8461
1-13	72	no	146.0341	1.2619	0.2846	23.3061	7.5611	33.3918
1-13	72	no	142.3538	1.2522	0.372	24.8359	7.9639	33.7677
1-14a	72	no	93.014	1.0903	0.0729	46.3431	24.2313	19.8416
1-14a	72	no	116.7176	0.6386	0.1422	24.6448	7.1184	45.0416
1-14c	72	no	192.5862	3.7735	1.2879	25.1962	6.1937	14.9647

TABLE 6-continued

Shake Flask Results								
Time Baffled			Fermentation Broth Analyte (g/L)					
Strain	Point	Flask	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-14c	72	no	188.6326	3.8733	1.2983	26.1231	5.7399	14.0219
1-8a	72	yes	85.7327	2.3671	0	64.5607	50.4989	0.3506
1-8a	72	yes	84.8898	2.3079	0	65.2192	50.0771	0.3258
1-8b	72	yes	51.5689	5.4855	0	65.6227	59.0813	2.8692
1-8b	72	yes	43.6927	5.1612	0	68.9601	65.7296	0.2363
1-8c	72	yes	145.4187	3.8764	0.2167	42.7845	26.0502	3.2431
1-8c	72	yes	151.2918	3.4321	0.2388	40.3376	24.5156	2.5922
1-13	72	yes	116.7837	2.1609	0	47.3081	36.4069	1.4031
1-13	72	yes	109.7339	2.1555	0	48.744	39.4191	1.5179
1-14a	72	yes	98.9757	1.5879	0	52.8139	35.1409	4.7515
1-14a	72	yes	89.9746	1.5549	0	54.1916	37.243	6.0006
1-14c	72	yes	147.7416	3.6853	0.4986	40.2659	23.9122	2.6688
1-14c	72	yes	142.0211	3.7815	0.473	40.3568	23.2301	5.5432
1-8a	96	no	42.8644	8.6314	1.9033	66.7209	16.8236	35.6326
1-8a	96	no	43.306	8.7595	1.8279	67.2738	16.6128	36.069
1-8b	96	no	42.576	3.5921	0.7858	48.1976	15.0165	47.2719
1-8b	96	no	39.3759	3.6963	0.8457	50.5586	15.2643	47.3769
1-8c	96	no	121.1279	6.1571	1.5101	51.1161	7.7732	24.5364
1-8c	96	no	119.7203	5.8701	1.5268	49.7403	7.4936	24.1627
1-13	96	no	86.8157	1.7625	0.5887	26.0683	8.171	49.5427
1-13	96	no	81.3414	1.8161	0.5429	28.1024	8.695	51.9138
1-14a	96	no	67.2068	1.3508	0.1064	54.414	23.4538	18.5858
1-14a	96	no	68.7416	0.8149	0.2827	25.286	7.6839	57.539
1-14c	96	no	126.5677	8.4837	2.5892	46.5038	7.8006	20.4329
1-14c	96	no	123.4764	8.2189	2.5542	47.1131	7.3021	20.5738
1-8a	96	yes	53.3442	2.8951	0	80.6748	54.7071	0.2221
1-8a	96	yes	51.6395	2.8795	0	82.5517	54.7041	0.1745
1-8b	96	yes	29.5099	6.3734	0	77.4994	64.3626	1.7047
1-8b	96	yes	24.9716	5.5745	0	76.997	69.4787	0.1281
1-8c	96	yes	91.2815	5.298	0.3619	67.0619	34.1619	1.2364
1-8c	96	yes	92.6441	5.1885	0.2913	67.3122	34.9422	0.474
1-13	96	yes	74.1743	2.8581	0	66.9987	45.3131	0.21
1-13	96	yes	69.5317	2.8072	0	67.201	47.9707	0.2132
1-14a	96	yes	62.1225	1.9868	0	68.7981	40.2958	2.7787
1-14a	96	yes	58.7857	1.9428	0	69.8722	42.3314	4.2523
1-14c	96	yes	98.3003	5.1312	0.5472	59.6042	27.5231	4.4645
1-14c	96	yes	95.5057	5.3736	0.5769	64.4065	32.0783	1.0075

Example 4—Shake Flask Fermentation Assay

[0085] Strains 1-14c, 1-15a-d, 1-16a-d, 1-17a-e, 1-18a-e, and 1-19a-e (outlined in Tables A and B above), were run in shake flasks to assess xylitol, erythritol, ribitol, glycerol, and ethanol production and glucose consumption.

[0086] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0087] A 250 ml non-baffled flask containing production medium (Table C) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 96 hours of incubation. Samples were analyzed for glucose, arabinose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance

liquid chromatography with refractive index detector. Fermentation results are reported in Table 7 and FIG. 4.

[0088] As outlined in Table 2, strain 1-14c was transformed with one of the recited XDH homolog candidates. Shake flask characterization of these strains demonstrated that none of the tested XDH homolog candidates increased xylitol production above the production level of the 1-14c parent strain under the same conditions. In fact, some strains had decreased xylitol production relative to the parent strain. **[0089]** While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the sequences was not correctly targeted to the ER1 locus. Further analysis indicated that strains 1-15a-d, 1-16a-d, 1-17a-e, 1-18a-e, and 1-19a-e include the transformed polynucleotide sequence, but it is not targeted to the ER1 locus.

TABLE 7

96 hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-14c	99.4	9.2	3.3	50.2	8.0	31.0
1-14c	97.0227	9.5367	3.1468	51.6515	7.91	31.5191
1-15a	124.5137	6.615	2.1213	53.0056	6.7519	26.1911
1-15b	176.5	3.8	1.4	41.1	6.8	14.3

TABLE 7-continued

96 hour Shake Flask Results						
Fermentation Broth Analyte (g/L)						
Strain	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-15c	158.5056	8.3277	1.9897	46.3165	6.3899	16.2085
1-15d	169.4034	7.4558	1.9438	40.7081	6.8775	16.2906
1-16a	183.2909	5.7001	1.8513	37.6993	6.532	13.9718
1-16b	195.2726	5.9215	2.0525	34.291	7.8959	10.4399
1-16c	186.0142	3.0993	1.2395	38.5607	6.4959	13.3875
1-16d	220.4559	4.0252	1.5282	26.3763	5.2339	8.364
1-17a	93.7651	7.9426	2.1475	61.8649	10.1967	30.4088
1-17b	109.4079	7.1854	2.0227	56.4907	8.9673	29.4155
1-17c	151.9165	7.2493	2.0357	47.4785	7.341	19.2757
1-17d	155.3155	5.1622	1.6969	46.4501	7.0876	19.89
1-17e	155.6752	9.3248	2.9191	45.036	7.2426	19.2712
1-18a	72.4171	5.9224	2.1062	65.4266	9.9271	35.9972
1-18b	91.4329	9.5635	3.1593	55.0515	8.7798	32.4985
1-18c	127.842	7.0101	2.5856	47.4543	7.4967	26.2408
1-18d	91.8716	6.3103	2.2494	57.7429	7.5496	36.6871
1-18e	80.0458	6.3959	2.164	60.4785	9.5805	37.3784
1-19a	89.4024	8.9136	3.0074	55.2153	8.9666	34.287
1-19b	114.2333	5.5967	1.5941	60.2268	8.8782	27.7048
1-19c	157.0952	7.9909	2.2512	46.2085	7.5793	17.9777
1-19d	85.8555	3.6704	1.2423	67.7445	9.3857	35.5171
1-19e	166.3217	4.1078	2.2277	40.8836	5.8864	21.8067

Example 5—Shake Flask Fermentation Assay

[0090] Strains 1-14c, 1-20a-e, 1-21a-e, 1-23a-e, and 1-24a-e (outlined in Tables A and B above), were run in shake flasks to assess xylitol, erythritol, ribitol, glycerol, and ethanol production and glucose consumption.

[0091] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0092] A 250 ml non-baffled flask containing production medium (Table C) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 96 hours of incubation. Samples were analyzed for glucose, arabinol, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 8 and FIG. 5.

[0093] As outlined in Table 2, strain 1-14c was transformed with one of the recited XKS/sugar phosphatase homolog candidates. Strains expressing the *S. cerevisiae* DOG2 sugar phosphatase, *A. bacterium* phosphatase homolog, or the *P. puerhi* HAD-1A family hydrolase homolog had increased final titers of xylitol titers. Results for the *Mesorhizobium* sp. HAD family hydrolase homolog candidate were similar to those reported for the *E. coli* hexitol phosphatase gene (data not shown).

[0094] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the

sequences was not correctly targeted to the ER1 locus. Further analysis indicated that strains 1-20a-e, 1-21a-e, 1-23a-e, and 1-24a-e include the transformed polynucleotide sequence, but it is not targeted to the ER1 locus.

TABLE 8

96 hour Shake Flask Results						
Fermentation Broth Analyte (g/L)						
Strain	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-14c	99.4	9.2	3.3	50.2	8.0	31.0
1-14e	97.0227	9.5367	3.1468	51.6515	7.91	31.5191
1-20a	109.8471	5.3636	3.5797	40.621	7.616	33.8712
1-20b	88.5895	8.2061	3.4587	50.5238	8.2101	35.4361
1-20c	194.6963	5.1945	3.8179	21.9057	3.667	14.9874
1-20d	158.3447	6.5005	3.7387	33.7401	4.7477	22.013
1-20e	121.7546	5.3639	2.6149	49.5375	6.4906	29.9388
1-21a	81.7409	11.0912	3.4072	55.3219	9.1535	34.8535
1-21b	79.2342	11.539	4.2206	28.6582	9.8801	47.4776
1-21c	95.6748	3.4423	1.23	57.8996	9.8309	37.3651
1-21d	134.4472	7.3051	2.9165	40.3628	7.1966	28.6082
1-21e	71.2417	8.4875	2.281	70.1432	9.6048	37.6368
1-23a	99.7368	11.7962	3.9377	55.9439	8.4156	30.7757
1-23b	98.576	6.8823	4.4492	42.9345	8.5033	40.8846
1-23c	105.586	9.9521	3.2203	52.3807	8.7781	32.1227
1-23d	86.3296	3.0776	2.7847	52.2668	11.9752	39.491
1-23e	88.5566	8.441	3.1327	56.6528	7.5047	36.5921
1-24a	98.2495	2.8919	1.4898	59.1482	7.474	32.5949
1-24b	115.6181	8.1052	2.586	47.4175	7.1073	29.597
1-24c	120.0083	9.3171	3.1328	52.7683	5.2785	25.0334
1-24d	92.7436	3.1304	1.5755	63.0015	7.9723	34.7408
1-24e	115.923	9.2322	3.3841	52.8348	7.1139	25.719

Example 6—Shake Flask Fermentation Assay

[0095] Strains 1-1, 1-20b, 1-24e, 1-25a-d, 1-26a-e, 1-27a-e, and 1-28a-c (outlined in Tables A and B above), were run in shake flasks in duplicate to assess xylitol, erythritol, ribitol, glycerol, and ethanol production and glucose consumption.

[0096] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0097] A 250 ml non-baffled flask containing production medium (Table C) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 96 hours of incubation. Samples were analyzed for glucose, arabinol, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 9 and FIG. 6.

[0098] The strains tested in this example include the phosphatase candidates that produced increased xylitol titers in Example 5 transformed into the wild-type background to observe the metabolite production. Strains expressing either

the *S. cerevisiae* DOG1 or the *S. cerevisiae* DOG2 sugar phosphatase enzymes produced more xylitol than the wild-type parent background.

[0099] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the sequences was not correctly targeted to the ER1 locus. Further analysis indicated that strains 1-20b, 1-24e, 1-25a-d, 1-26a-c, 1-27a-c, and 1-28a-b include the transformed polynucleotide sequence, but it is not targeted to the ER1 locus.

TABLE 9

Average 96 hour Shake Flask Results						
Fermentation Broth Analyte (g/L)						
Strain	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-1	65.84	0.54	0.17	53.37	16.13	40.53
1-20b	72.25	3.68	5.10	28.32	8.51	51.17
1-24e	93.81	7.96	4.15	41.51	7.66	35.83
1-25a	110.33	0.55	0.20	34.05	16.34	34.64
1-25b	57.89	3.83	2.00	46.75	13.08	48.28
1-25c	30.80	1.18	0.58	52.12	24.52	50.73

TABLE 9-continued

Average 96 hour Shake Flask Results						
Fermentation Broth Analyte (g/L)						
Strain	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-25d	88.74	0.65	0.70	40.08	20.89	38.58
1-26a	103.26	1.21	2.62	31.62	10.16	39.15
1-26b	104.19	0.92	1.82	34.26	11.94	38.68
1-26c	100.23	2.34	4.40	25.45	6.34	43.85
1-26d	47.48	0.95	1.33	64.81	10.16	46.51
1-26e	83.76	0.80	1.97	34.24	11.14	45.99
1-27a	65.19	0.68	0.19	58.14	16.17	40.48
1-27b	62.19	0.66	0.19	55.95	16.20	43.71
1-27c	68.36	0.67	0.18	56.33	15.43	41.59
1-27d	86.03	0.48	0.19	45.27	15.81	39.76
1-27e	98.16	0.33	0.15	37.02	13.95	38.39
1-28a	94.61	0.63	0.19	48.79	14.47	35.63
1-28b	68.15	0.75	0.15	56.16	13.31	41.42
1-28c	78.41	0.75	0.15	54.84	21.73	38.09

SEQUENCE LISTING

Sequence total quantity: 101

SEQ ID NO: 1 moltype = AA length = 363
 FEATURE Location/Qualifiers
 source 1..363
 mol_type = protein
 note = Formerly *Pichia stipitis*; referred to as *Pichia*
 stipitis in the priority application and the present
 application
 organism = *Scheffersomyces stipitis*

SEQUENCE: 1
 MTANPSLVLN KIDDISFETY DAPEISEPTD VLVQVKKGTI CGSDIHFYAH GRIGNFVLTK 60
 PMVLGHESAG TVVQVGKGV TSLKVGDNVAI EPGIPCRFCD ECKSGHYNLC PHMAFAATPN 120
 SKEGEPNPPG TLCKYFKSPE DFLVKLPDHF SLELGALVEP LSVGVHASKL GSVAFGDYVA 180
 VFGAGPVGLL AAATAKTFGA KGVIVVARSD RKLKMAKDIG AATHTFNSKT GGSEELIKAF 240
 GGNVNVVLE CTGAEPICIKL GVDAIAPGGR FVQVGNAAGP VSPFITVFAM KELTLFGSFR 300
 YGFNDYKTAV GIFDTNYQNG RENAPIDFEQ LITHRYKFKD AIEAYDLVRA KGKAVKCLID 360
 GPE 363

SEQ ID NO: 2 moltype = AA length = 550
 FEATURE Location/Qualifiers
 source 1..550
 mol_type = protein
 organism = *Hypsizygus marmoreus*

SEQUENCE: 2
 MSSQVSGPLF LGLDLSTQQL KAILISEDHT VVHESAVHFD NDLPSYGTN GAIRGPGVGE 60
 VTSPVAMWLE AIDLLMDRMK KADVKLDAIS AVSGAGQOHG SVYWSNDAAE LLSSLDPAQS 120
 LVSQLSPGAF SLPQAPIWQD SSTTQECRDL EQAVGGPQVL ADITGSRAYE RFTGNQIAKI 180
 RRLHPESYNA TTRISLVSSF IPSVFLGYIA PIDTSDASGM NLMNVLTCKW DDDLLNACGG 240
 EMLRSKLGPE PVIGGTVLGK VHQWVVERWG LNPDCIVAPF TGDNPASVMA LSAPGDAILS 300
 LGTSTTFLLS IPPADTPPQR FTTSHLLSH TDLDTIAML CYKNGALARE QVRDHNADGH 360
 WTRFNLVEQ TPPGNDGYIG LYFPLPEIIP PGVVGQHAFH VESLDNTAPL AVKDLPEKAH 420
 PRAILESQFL SIRSRIADIL PEHAPHLKRL VATGGSSANQ TIRQLAADIF DMKVFISTTK 480
 EAAGMGGALL AKYSWWKTQH TNGRTFEDMT GGEITGLQCV AEPRGSVTGV YDTLVGVYRA 540
 CEEQVVIKCS 550

SEQ ID NO: 3 moltype = AA length = 553
 FEATURE Location/Qualifiers
 source 1..553
 mol_type = protein
 organism = *Cylindrobasidium torrendii*

SEQUENCE: 3
 MIENPTFLGL DLSTQQLKAV LVQEDCSIVH ESAVRFEEDL PKYGTCKGAH KPGDGVVTSP 60
 VAMWLDALDL LMEKMRKAGV DFASIAAISG AGQOHGSVYV SNAEECLQA LDASKPLTQL 120
 APAAFSLQTS PIWQDSSTTR ECREIEEAIG GAQALADISG SRAYERFTGT QIKKLRRVNR 180

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AAYDSTSRVS	LVSSWIPSIF	LGTVAPIEIS	DASGMNLM DV	ISCKWDDRLL	EICGGPELRS	240
KLGPPEVPGG	TSLGTVCWW	VERWGFSP	IVAPFTGDN	ASVAALSLPG	DALLSLGTST	300
TYLLSIPPAS	EAPKRFTTSH	LLSHPTTAPD	AQIAMLCYKN	GALAREQVRD	RFADADWSKY	360
NDAVEQTHPG	SEKYMGLYFP	LPEIIPPGVK	GSFVYRTEMG	KKPVKVEEKD	VPTLHPRMI	420
LESQFLSIRS	RIAAILPPNS	PPLQRLVVTG	GSSANQTIRQ	LAADLFDMKV	YVSATKEAAG	480
MGGALLARYA	WWRATEDDMG	TFEEMCAGLG	SEALGMKCVA	EPNHAAALEY	AQLVEPYVAC	540
EAEVVRECNA	HLP					553

SEQ ID NO: 4 moltype = AA length = 549
 FEATURE Location/Qualifiers
 source 1..549
 mol_type = protein
 organism = Pleurotus ostreatus

SEQUENCE: 4

MAATSSQPLF	LGLDLSTQQL	KAVVITQDLD	IVHESAVNFA	RDLPOYGTEN	GAIKGPDEGE	60
VTSPVEMWVE	AMDLLLARMK	EASVDFAAIV	AVSGAGQQHG	SVFWSKDAES	RLSSLDPSRT	120
LLEQLSPTAF	SLMNSPIWQD	SSTTKQCLEL	EESIGGPQAL	ADLTGSRAYE	RFTGTQIAKI	180
HQTRPEVYAP	TSRISLVSSF	IPSLFLGSVA	PIEVSDASGM	NLMNVLTCKW	DDRLLLEACGG	240
SALRSKIGPE	PVSGGSNLGK	VHKWWVERWG	FQPD CIVAFV	TGDNPATVVS	LSAPGDALLS	300
LGTSTTFLLS	IPPADTPPKR	FTTSHLLSHP	TNLDAQIGML	CYKNGALARE	QIRDRYADND	360
WGKFNTLLEQ	APVGNNGLLG	FYFPLPEIIP	PGVVGEFFYN	VDGATPALVE	GLPDNAHPRA	420
ILESQFLSIL	SRVEAILPEH	APHLKRLVLT	GGSSANQTIR	QLAADLFGMK	AYVSATKEAA	480
GLGGAMLAKY	AWWKQINGGK	GTFEAMTGE	MPGMT CVAEP	RDEVTQIYEG	LVS VYRLCEE	540
DVVSRSQRK						549

SEQ ID NO: 5 moltype = AA length = 566
 FEATURE Location/Qualifiers
 source 1..566
 mol_type = protein
 organism = Trametes coccinea

SEQUENCE: 5

MLDDPLFLGL	DLSTQGLKAV	LIDADGAPVH	ESSVNFDRDL	PSYGTNGAI	RGPGAGEVTS	60
PVRMWLDAYD	LIMERMRSAG	VDFSRLILGIS	GDGQQHGSVY	WSTEAERLLA	SLDSSRPLAE	120
QLSPAAFSIQ	RAPIWQDSST	TRECRELEAA	IGGAQALADL	TGSRAYERFT	ANQIAKIRRT	180
DPDSYRATAR	ISLVSSFMPS	LFLGSIAPIE	VSDASGMNLM	NVLTCWDDRR	LLEAAGGPPEL	240
RAKIGPEPAP	GGTFLGKVHP	YWVQRWGFNP	ECGVAPFTGD	NPATVVALST	PGDAILSLGT	300
STTLLLSIIP	AETAPKRFTT	SHLLSHPTTP	DAHIAMLCYK	NGALAREQVR	DQYADGHWTR	360
FNELVEARPV	GNEGVMGYFY	PLPEIIPPNV	QGNFFFSTSP	SPSPSSPSSQ	SSITALPEAS	420
MPRPAHPRAI	LESQFLSIRS	RITAILPPDA	PPLQRLVLSG	GSSRNPIVQQ	AAADVFGMRV	480
FVAGAEAGAAE	GGAQLARYAR	WRARNGGVGS	FEEMRAMDGE	GEGEGGERRR	CVAEPKWERE	540
RMYAGLVETY	RACEDLVVQM	CERAVA				566

SEQ ID NO: 6 moltype = AA length = 551
 FEATURE Location/Qualifiers
 source 1..551
 mol_type = protein
 organism = Antrodiaella citrinella

SEQUENCE: 6

MSTDTPPLFL	GLDLSTQGLK	AVLITEDSQV	VHEASVRFQD	DLPQYGTVNG	AVRGPGEDEV	60
TSPVKMWLEA	IDLLMQRIKA	TGVQTSRILG	VSGDGQQHGS	VFWSSDAENL	LSTMNPEQGL	120
ADQLAPAAFT	LPNAPIWQDS	STTRECRELE	EAVGGPQALA	DLTGS RAYER	FTGTQIRKIK	180
RVNPEAYNAT	SHIALVSSFL	PSLFLGAIAP	IEISDASGMN	LMNVLT YKWD	EKLDDVAGGY	240
ELRSKLGPEP	VPGGVVLGKV	SEWVWKRWGF	NPD CIVAPFT	GDNPATVVAL	SSPGDAILSL	300
GTSTTLLLSI	PPSDTAPKRF	TSSHLLAHP	TYDAHIAMLC	YKNGALARDQ	VRNKYANADW	360
AKFNELVEQT	PTGNNLGLGY	YFSLPEIIPP	NVVGNYFYST	SPSIRAVPES	EVSAAHPRA	420
VLESQFLSIR	SRIQAVLPAN	SPPLQRLVLS	GGSSSNPIIR	QLAADVFGMR	VFVAGQEGAA	480
EGGAQLARFA	WWEENGGRG	TFEEMRAFDG	EEERMKCVAE	PKEDNVRYVA	GLVDAYRECE	540
ERVVELCAQT	A					551

SEQ ID NO: 7 moltype = AA length = 628
 FEATURE Location/Qualifiers
 source 1..628
 mol_type = protein
 organism = Ustilago maydis

SEQUENCE: 7

MEGTSPPFFLG	LDASTQALKA	SLLDVDLNLV	SELEIRFDRD	LPQYKTTGGI	SAPAADDQDG	60
TVVAPVMLYV	ESIDMLAERM	RSASWPLSRI	RAISAAGQOH	ASVYFSRAAP	RIFTTLSAEK	120
TLTEQVEQAF	SRKVVPINWQD	SSTVDACKAF	EKAVGGA EKL	AQMTGSKAHT	RFTGPQIYKF	180
RTQQPQAYKD	TERIGLVSSF	ITTLCLVGGQ	EDESSVIKGI	DEADACGMNL	LDMRPASSIT	240
ATKGRIEPCW	NQTLALALASG	ESEGS DASLG	GAVELERKLG	SIYRDAGTPV	AKIGS WWTCK	300
YGFSPDCHVF	PGTGNPATF	LAFSLAEREA	IISLGTS DTV	MVATHQFVPD	PDFHAFPHPA	360
RLPSSGDAPF	NMLVYKSGSL	AREWIRDQYC	NSDWA AFNAD	VEKFKLQDDG	EQRAGFYWLR	420
PEIIPPGASG	IYRIKPTSC	GEWQKVDDFS	DKGMNASGIL	ETQMLNRYR	SSSIVAGPTT	480
SSGS AVDTSV	LPLQAIYAVG	GASSNPITITQ	TMADVFGCDI	VKPVQPAQDG	KEWTPANYNF	540
CSVGAAYKAA	WGSRLQLKA	GEKVPEFDEF	VHTTKSSQAK	RAGSVREGTG	EVKGSVQVIC	600
RPTGRTQVY	TNVIGAWKQL	EERAQKEQ				628

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SEQ ID NO: 8 moltype = AA length = 246
 FEATURE Location/Qualifiers
 source 1..246
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 8
 MAEFSADLCL FDLDTIVST TVAAEKAWTK LCYEYGVDPD ELFKHSHGAR TQEVLRFFFP 60
 KLDDTDNKGV LALEKDIAHS YLDTVSLIPG AENLLLSLDV DTETQKKLPE RKWAIVTSGS 120
 PYLAFSWFET ILKNVGKPKV FITGFDVKNG KPDPEGYSRA RDLLRQDLQL TGKQDLKYVV 180
 FEDAPVGIKA GKAMGAIIVG ITSSYDKSVL FDAGADYVVC DLTQVSVVKN NENGIVIQVN 240
 NPLTRA 246

SEQ ID NO: 9 moltype = AA length = 505
 FEATURE Location/Qualifiers
 source 1..505
 mol_type = protein
 organism = *Bacteroides thetaiotaomicron*

SEQUENCE: 9
 MYILAHDLGT SGNKATLFDE SGLLIASRTA AYPTDYASGN RAEQNPHHWW KAIVDTTQAL 60
 LELVSPNDIA GVALSGQMMG CLCVDKDGNP LRPMLLYCDQ RSQEEEAHLT EKIDPLHFYE 120
 ITGHRISASY SVEKLMWVKK HEPEIFAQTA KMLNAKDYIN YRLCGTIATD PSDASGTNAY 180
 DLNRWQWSEE IIEAAELDLS LPFEVRSSID VIGEITNEAA RETGLLAGTP VICGGGDGSC 240
 AGVGVGCVAP GTAYNYLGSS SWVALTVEKP IVDEQRRTMN WAHVVPGLMH PSGTMQAAGS 300
 SYNWMINQLC QNEQALAAQS GRSVFEELIDE QIIASPIGAN KLLPLPYMLG ERTPRWNVDA 360
 KGAFIGLTLG HKHGDMLRAV MEGITLNLGF IINIPRKHVP IDRMTVIGGC AQNPVVRQMM 420
 ADIYQAEIRV PNYLEEATSM GAAILAGIGA GVFPDFSVID RFVRIEQTVQ PIPENVKKYE 480
 AWMPVFDQAY HALCEMYTEI VKTEL 505

SEQ ID NO: 10 moltype = AA length = 623
 FEATURE Location/Qualifiers
 source 1..623
 mol_type = protein
 note = Formerly *Pichia stipitis*; referred to as *Pichia*
 stipitis in the priority application and the present
 application
 organism = *Scheffersomyces stipitis*

SEQUENCE: 10
 MTTTFFDAPD KLFLGFDLST QQLKIIVTDE NLAALKTYNV EFDSSINSSVQ KGVIAINDEI 60
 SKGAIISPVY MWLDALDHVF EDMKKDGFPP NKVVGISGSC QQHGSVYWSR TAEKVLSELD 120
 AESSLSSQMR SAFTFKHAPN WQDHSTGKEL EEFERVIGAD ALADISGSRA HYRFTGLQIR 180
 KLSTRFKPEK YNRTARISLV SSFVASVLLG RITSIEEADA CGMNLVDIEK REFNEELLAI 240
 AAGVHPELDG VEQDGEIYRA GINELKRLKG PVKPITYESE GDIASYFVTR YGFNPDCIKY 300
 SFTGDNLATI ISLPLAPNDA LISLGTSTTV LIITKNYAPS SQYHLFKHPT MPDHYMGMIC 360
 YCNGSLAREK VRDEVNEKFN VEDKKSWDKF NEILDKSTDF NNKLGIFYPL GEIVPNAAAQ 420
 IKRSVLNSKN EIVDELGDGK NWQPEDDVSS IVESQTLSCR LRTGPMLSXS GDSSASSSAS 480
 PQPEGDGTDL HKVYQDLVKK FGDLYTDGKK QTFESLTARP NRCYVVGAS NNGSIIRKMG 540
 SILAPVNGNY KVDIPNACAL GGAYKASWSY ECEAKKEWIG YDQVINRLF VSDMNSFEV 600
 KDKWLEYANG VGMLAKMESE LKH 623

SEQ ID NO: 11 moltype = AA length = 600
 FEATURE Location/Qualifiers
 source 1..600
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 11
 MLCSVIQRQT REVSNMTSLD SYLGFDDLST QQLKCLAINQ DLKIVHSETV EFEKDLPHYH 60
 TKKGVYIHGD TIECPVAMWL EALDLVLSKY REAKFPLNKV MAVSGSCQOH GSVYWSSQAE 120
 SLLEQLNKKP EKDLLHYVSS VAFARQTAPN WQDHSTAKQC QEFEECIGGP EKMAQLTGSR 180
 AHFRPTGPQI LKIAQLEPEA YEKTKTISLV SNFLTSLVG HLVELEEADA CGMNLVDIRE 240
 RKFSDELLHL IDSSSKDKTI RQKLMRAPMK NLIAGTICKY FIEKYGFNTN CKVSPMTGDN 300
 LATICSLPLR KNDVLVSLGT STTVLLVTDK YHPSPNYHLF IHPTLPNHYM GMICYCNGSL 360
 ARERIRDELN KERENNYEKT NDWTLFNQAV LDDSESENE LGVYFPLGEI VPSVKAINKR 420
 VIFNPKTGMI BREVAKFDDK RHDAKNIVES QALSCRVRIS PLLSDSNASS QQLNEDTIV 480
 KFDYDESPLR IDLNKRPERT FFFVGASKND AIVKKFAQVI GATKGNFRLE TPNSCALGGC 540
 YKAMWSLLYD SNKIAVFPDK FLNDNFPWHV MESISDVNDW NWDRYNSKIV PLSELEKTLI 600

SEQ ID NO: 12 moltype = AA length = 272
 FEATURE Location/Qualifiers
 source 1..272
 mol_type = protein
 organism = *Streptomyces avermitilis*

SEQUENCE: 12
 MARPTVLVAS DLDRTLIYSA AALALTMPDA RAPRLLCVEV HENKPLSYMT ETAAGLLTEL 60
 GDAAVFVPTT TRTRKQYQRI NLPGPVPKYA ICTNGGQLLV DGVDPDPDWH VVTGRLADEC 120
 APLDEVQAH L AETADPVWLR KQVVAEDLFV YLVVERELLP EAWVKELAVW AENRGWTVSL 180

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QGRKLYAVPK PLTKSAAMRE VARRTGAGLT LAAGDSLDDA DLLLAADRGW RPHGHELADA 240
EWAAPAVAAAL SERGVLAGER ILREFLRGVR EG 272

SEQ ID NO: 13 moltype = AA length = 256
FEATURE Location/Qualifiers
source 1..256
 mol_type = protein
 note = Rhodobacteraceae bacterium
 organism = unidentified

SEQUENCE: 13
MYAELFRLDG RTAFVTGGSR GIGLACAEAL GEAGATVALS SRSRDEGEAA AAALQGKGIR 60
AIYLPADLST EKAAQEAVER ADELGGGLDI LVNNAGIARH CDSLALPPET WDEVIATNLT 120
GLFWCCRAAL ERMVAAGRGS IVNVGSISGF ISNLPQNQVA YNASKASVHM LTKSLAGEFA 180
QQNIRINAVA PGYIETAMTR GGLDDPEWSK IWLGMTPMRR SGTPSEVAAA VLFLASDAAS 240
YITGSVLTID GGYTIH 256

SEQ ID NO: 14 moltype = AA length = 352
FEATURE Location/Qualifiers
source 1..352
 mol_type = protein
 organism = Bemisia argentifolii

SEQUENCE: 14
MASDNLSAVL YKQNDLRLEQ RPIPEPKEDE VLLQMAYVGI CGSDVHYEH GRIADFIVKD 60
PMVIGHEASG TVVVKGNVVK HLKKGDRVAV EPGVPCRRQC FCKEGKYNLC PDLTFCATPP 120
DDGNLARYYV HAADFCHKLP DNVSLEEGAL LEPLSVGVHA CRRAGVQLGT TVLVIGAGPI 180
GLVSVLAAGA YGAFVVCAR SPRRLEVAKN CGADVTLVVD PAKEEESII ERIRSAIGDL 240
PNVTIDCSGN EKCITIGINI TRTGGTLMV GMGSMQMTVP LVNACAREID IKSVFRYCN 300
YPIALEMVAS GRCNVKQLVT HSPKLEQTV D AFEAARKKAD NTIKVMISCR QG 352

SEQ ID NO: 15 moltype = AA length = 356
FEATURE Location/Qualifiers
source 1..356
 mol_type = protein
 note = Bos indicus x Bos taurus
 organism = Bos sp.

SEQUENCE: 15
MAAAKPENLS LVVHGPGLDL LENYPIPEPG PNEVLLKMHS VGICGSDVHY WQHGRIGDFV 60
VKKPMVLGHE ASGTVVVGVS LVRHLQPGDR VAIEPGAPRE TDEFCKIGRY NLSPTIFFCA 120
TPDDGNLRC FYKHANFCY KLPDNTVFEE GALIEPLSVG IHACRRAGVT LGNKVLVCGA 180
GPIGLVSLLA AKAMGAQVQV VTDLSASRLS KAKEVGADFI LQISNESPOE IAKKVEGLLG 240
SKPEVTIECT GVETSIQAGI YATHSGGTLV LVGLGSEMTS VPLVHAATRE VDIKGVFRYC 300
NTWPMASML ASKSVNVKPL VTHRFPLEKA LEAFETSRKG LGLKVMIKCD PNDQNP 356

SEQ ID NO: 16 moltype = AA length = 358
FEATURE Location/Qualifiers
source 1..358
 mol_type = protein
 organism = Aspergillus tubingensis

SEQUENCE: 16
MSTQNTSAQN LSFVLEGIHR VKFEDRPIPE IKDPHDVLVN VRFTGICGSD VHYWEHGSIG 60
QFVVKDPMVL GHESGVSISK VGSVAVSLKV GDCVAMEPGI PCRRCEPCKA GKYNLCVKMA 120
FAATPPYDGT LAKYYVLPED FCYKLPESIT LQEGALMEPL SVAVHIVKQA GINPGQSVV 180
FGAGPVGLLC CAVAKAYGAS KVIAVDIQKG RLEFAKKYAA TATFEPAKAA ALENAQRLIA 240
ENDLGSGADV AIDASGAEPS VHTGIHVLRA GGTYYVQGGMG RSEITFPIMA ACTKELNVKG 300
SPRYGSGDYK LAVSLVSGSK VNVKELITGV VKFEDAERAF EEVKAGKGIK TLIGGVDS 358

SEQ ID NO: 17 moltype = AA length = 383
FEATURE Location/Qualifiers
source 1..383
 mol_type = protein
 note = Chaetomium thermophilum var. thermophilum DSM 1495
 organism = Chaetomium thermophilum

SEQUENCE: 17
MAAGKSNLSF VLKKNPNEVS EERPVPKLQS PHDVLVAVNY TGICGSDVHY WVHGAIGHFV 60
VKDPMVLGHE SAGTVVEVGS AVTSLKPGDR VALEPGYPCR RCNNCLSGKY NLCPDVVF 120
TPPYDGTLTG FWTAPADFCY KLPDNLVSLQE GALIEPLAVA VHIVRQARVS PGQSVVVMGA 180
GPVGLLCAAV ARAFAGASKV SVDIVQSKLD FAKKFAATHT YMSQSVSAEE NAKNLLKAAD 240
LPGGADVVID ASGAEPSTQT SLHVVRMGGT YVQGGMGKAD ITFPIMALCL KEVTARGSPR 300
YSGSDYKLA ELVSGSKVDV KQLINGVVPF EKAEDAFKKV REGQVIKILI AGPNEKVENG 360
GLDTTVDEKK LKEAKVNGST GCC 383

SEQ ID NO: 18 moltype = AA length = 344
FEATURE Location/Qualifiers
source 1..344
 mol_type = protein
 note = Candidatus Poribacteria bacterium

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                                organism = unidentified
SEQUENCE: 18
MKAAYLVHKVR DIRIEELPKP QIVNDDDLV RVKAVGVC GS DIHFYMHGRI GDFVVEKPLI 60
LGHESAGIVE EVGKGVKDLK PGDKVALEPG IPCRRCEFC SGRYNLCPHV RFFAAPPVNG 120
VFAEYVVHPA DFCFKLPENA SLEEGAMVEP LSVGLHGAIL GGVRPGDWVA VLGSGPIGLM 180
ALQASLAMGA SRVICVDLYD FRLKLAKELG ASETINAKET DPVEAIADLT GGRGVDVVF 240
AAGVPKTSQQ AIEIARPGGT IVHIGLGEQE AVPMNIVKMI FKELRLQGIH RYANIYPRAI 300
KLIASGRVKV KPMITARLPL ERVEEALRMP VERPDSTIKV IVEI 344

SEQ ID NO: 19      moltype = AA length = 350
FEATURE           Location/Qualifiers
source            1..350
                  mol_type = protein
                  organism = Linepithema humile

SEQUENCE: 19
MAKDNLTAIL YGINDLRLN TNIEEPGDDE VLLEMGCVGI CGSDVHYLVN GRIGDFIVKK 60
PMIIGHESG TVAKLGKNVK NLEVGDRAI EPGVPCRTCT FCKEGRYNLC KDIVFCATPP 120
VHGSRRYYK HAADFCFKLP DHVSLEEGAL LEPLSVGVHA CKRGIGIDS KVLILGAGPI 180
GLVTLLVAKA MGASKVITD IVESRLKMAK KLGADNTYLV QKDRSEKDAI ADIHAIFGDE 240
PNRTIDASGA QASIRLAILA TKS GG VVVLV GNTAPEVQIP LINALVREVD IRGIFRYAND 300
YDDALNLVAS GKIDVKPLIT HNYKIEETK AFETSRTGAG GAIKVMHCK 350

SEQ ID NO: 20      moltype = AA length = 246
FEATURE           Location/Qualifiers
source            1..246
                  mol_type = protein
                  organism = Saccharomyces cerevisiae

SEQUENCE: 20
MPQFSVDLCL FDLGTIVST TTAESA WKK LCRQHGVDPV ELFKHSHGAR SQEMMKKFFP 60
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PYLAFS WFT ILKNVGKPKV FITGFDVKN KPDPEGYSRA RDLLRQDLQL TGKQDLKYV 180
FEDAPVGIKA GKAMGAI TVG ITSSYDKSVL PDAGADYVVC DLTQVSVVKN NENGIVIQVN 240
NPLTRD 246

SEQ ID NO: 21      moltype = AA length = 223
FEATURE           Location/Qualifiers
source            1..223
                  mol_type = protein
                  note = Acidobacteria bacterium
                  organism = unidentified

SEQUENCE: 21
MRTFLCSAIL FDLGVLVDS TRSVSRQWRQ WAGEHGIDPE KVLAI AHGVR TIEVVQGLAP 60
HLDLEAEVRR IERREAEDTD GVAVMPGAAD LVR SIPDGRW CVVTSGTRL L ATARLRLGGL 120
PMPGVLVTAD DVTNGKPDPE PYLKGAELLG VKPAECLVIE DAPAGIRAAH AGGMKVIALA 180
STYPVSELQE ADTVVRS LKS VAISAAAGTL EVRVAGEEME AAR 223

SEQ ID NO: 22      moltype = AA length = 210
FEATURE           Location/Qualifiers
source            1..210
                  mol_type = protein
                  organism = Mesorhizobium sp.

SEQUENCE: 22
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LPGVDPVVEA DALLKAEAAD LEGIVPIPGA VAF LKSLPPE RWAVVTSAPR ELALLRMAAA 120
GIPVPAIMVS AEDVTRGKPA PDCFLLA KR LGVEARECLV FEDAPAGIAA GEASGASVMV 180
ISATHVHPVV TQHASIATYD ALGVATDDNG 210

SEQ ID NO: 23      moltype = AA length = 191
FEATURE           Location/Qualifiers
source            1..191
                  mol_type = protein
                  organism = Escherichia coli

SEQUENCE: 23
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EADIAAEPT R LEHIEATETE GITALPGAIA LLNHLNKAGI PWAIVTSGSM PVARARHKIA 120
GLPAPEVFVT AERVKRGKPE PDAYLLGAQL LGLAPQECVV VEDAPAGVLS GLAAGCHVIA 180
VNAPADTPRL N 191

SEQ ID NO: 24      moltype = AA length = 214
FEATURE           Location/Qualifiers
source            1..214
                  mol_type = protein
                  organism = Puerhibacterium puerhi

SEQUENCE: 24
MRFDVEAVLF DIGTLVDST PAVERIWRW AGRRGLDAEE ILRVCHGRRT EDTLALFLPP 60
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SEQ ID NO: 25 moltype = DNA length = 4153
 FEATURE Location/Qualifiers
 source 1..4153
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 25

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SEQ ID NO: 26	moltype = DNA	length = 4159
FEATURE	Location/Qualifiers	
source	1..4159	
	mol_type = other DNA	
	organism = synthetic construct	

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SEQ ID NO: 27 moltype = DNA length = 4234
 FEATURE Location/Qualifiers
 source 1..4234
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 27

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 organism = synthetic construct

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SEQ ID NO: 29 moltype = DNA length = 4135
 FEATURE Location/Qualifiers
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 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 31      moltype = DNA length = 3579

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FEATURE
source
Location/Qualifiers
1..3579
mol_type = other DNA
organism = synthetic construct

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FEATURE
source
Location/Qualifiers
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mol_type = other DNA
organism = synthetic construct

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FEATURE            Location/Qualifiers
source              1..3483
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 34      moltype = DNA length = 3552
FEATURE            Location/Qualifiers
source              1..3552
                    mol_type = other DNA
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SEQ ID NO: 35      moltype = DNA length = 3968
FEATURE            Location/Qualifiers
source              1..3968
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 36 moltype = DNA length = 1949
 FEATURE Location/Qualifiers
 source 1..1949
 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 37 moltype = DNA length = 1476

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FEATURE
source
Location/Qualifiers
1..1476
mol_type = other DNA
organism = synthetic construct

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SEQ ID NO: 38
FEATURE
source
Location/Qualifiers
1..1276
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 38
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SEQ ID NO: 39
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source
Location/Qualifiers
1..4447
mol_type = other DNA
organism = synthetic construct

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FEATURE	Location/Qualifiers
source	1..4889
	mol_type = other DNA
	organism = synthetic construct

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 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 44 moltype = DNA length = 4928
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 mol_type = other DNA
 organism = synthetic construct

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moltype = DNA length = 4745

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Location/Qualifiers

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mol_type = other DNA
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FEATURE            Location/Qualifiers
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                    mol_type = other DNA
                    organism = synthetic construct

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source             Location/Qualifiers
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SEQ ID NO: 53      moltype = DNA length = 679
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                    organism = synthetic construct

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SEQUENCE: 53
agctcagacc ttgcattgta gccttgacct tgcctctatg accactaatt cgcacaagat 60
tgcaagttac catgtgcac ttgccatgta tccaatgttt cctgtttgcc atagtacgta 120
gtagatacat cgatcatcga tgcagatgac gcagatggca ttggttcgct ttgcctatg 180

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catccatcat	gcacatgca	tcatgcatca	tgcgtcacta	tccatattaa	tcattgcect	240
ctgtactagg	ccatgacacc	tgcagaccct	tcccagggga	tgttcttgt	aatagccaat	300
atactacaca	actttaccga	ttacgatcga	gatcgatctc	actatagttt	gtcgaacttc	360
gatgcacatc	gaggggtcgg	aaaaaagctg	acggaagcgg	aagtgggtga	attggttaat	420
ttagtttagat	tggcaggggt	gttcccaaga	ttcaacgcaa	gctgagaaag	aaaacagaaa	480
cacgagggcat	aagggccagtc	agaaaatgca	ttacgagggg	agttgaaagt	actgtggggc	540
agtgccctggg	tgcctcatgtg	caaatttttt	cttcgggttag	actgccgcag	catgcagcaa	600
ctctgccggc	cctatataac	tagtgcttca	tcttcttgtc	attcttgtct	ctttaccaac	660
acaccggtac	tctttcac					679

SEQ ID NO: 54 moltype = DNA length = 1002
 FEATURE Location/Qualifiers
 source 1..1002
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 54

catcgttcca	gcttgcttgg	gggtgtagcc	agcgcaaaag	tccgacgaac	agacgaattg	60
accgtagtcg	atgacagtcg	atgacagtcg	atgacagtcg	atgacagccg	acgaccgacg	120
aacaacaacc	aaccgcacgc	agcatgaggc	cgagaggctc	aggctgagcc	gctaaggcgc	180
tgaagcttga	agctcggttg	tccagtcaga	tctcatgata	tcacaccgcc	gcgctgtctc	240
ggtcgtatgg	ttcaatttgc	tgtgcctagc	ctagcctggg	tacgtcggtg	actagccact	300
agccactggc	cattatttgg	ctactgggca	cactaggagg	cgcactagtc	catcgcaagc	360
ggtaacttac	ctacccggca	ctacccgatg	atttccctct	tgcctggcgg	cctggcgggc	420
tggcggcgca	ctcttttgcg	tgggacccgc	aaattgcaaa	ccaacaaacg	actagtgtcc	480
ggtagctgtg	cccaagtaac	ctgggcagac	atagaaccac	attgatcgta	gggaaccgca	540
atggaccagt	acgctattct	cagtacattc	tgtacacttt	tgcctagatt	ggaatctggg	600
gaggatcaag	atcaaatctc	aaaccaactc	cacggcactg	gttcccgaca	cttggaactgt	660
agtagctgag	tactgaggtg	ctgaggtact	gaggctcaac	tgaagtctca	ctgaccgtga	720
cgcgcttctg	tgggtggcgt	ggattctagt	gttggatgtg	cgcagtggtg	tgtgctgggg	780
tgcgttcaga	ttcgtcggta	gaacttgact	attcacgtct	tgtgctgcgc	tacgctgtgg	840
aacctggaat	gtttaaaccct	gggtcctgag	ggtctccacc	cccgaagtcg	actgggttct	900
atataaagtc	gggtaggcaa	ttggctaggg	agaatctctc	ttttctcttt	ccccagaaca	960
cctgttcact	cactttctac	ctctttactt	acctacttca	ca		1002

SEQ ID NO: 55 moltype = DNA length = 1000
 FEATURE Location/Qualifiers
 source 1..1000
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 55

cctcgttctt	gattgcttcc	aaaagtaatt	taaggagaga	acctggctgt	ttgttgctct	60
actgaatgat	cttacctata	ccagcatacg	acaccttgct	gcccctcccc	ttcaccacca	120
ccatcaccac	tttctctctt	ttctttgggt	ttttctagtc	gttgcaattta	tggttgattc	180
gctcttttag	aacgctatgg	ctgtttattcc	ttcccaactc	ttttgtctgt	tttatcggtg	240
catccaaacg	tttgggctct	tcgtacctct	ttttctctgt	tttgctcttt	atttttgggt	300
tttctatccg	tttgatgggt	gtgactcttg	ttttcttttg	ggaagggttg	agctcaactt	360
ctcttcggtc	cttggtggag	tttgacttcc	ttttcttttt	cagctgtggc	cgatcattac	420
gatcaataat	tttgtaatat	gatatcgact	acatcctgaa	tcttttagcg	aaccatagca	480
gtttagcgtc	atagatatgc	atgcattgat	tacatcttac	acgcaccacg	gatgatgctc	540
agcgtctcag	cacgcagcgc	ctcagccgct	cgctcggtac	gctcggtcat	gatgtgcatt	600
atgcaagcct	aattgattgac	gggtacttga	tctgtgtccc	aattcgatcg	agatcgatct	660
cactatagtt	tgtcgaaact	cgatgcacat	cgaggggtgc	gaaaaaagct	gacggcagcg	720
gaagtgggtg	aattgggttaa	tttagttaga	ttggcagggt	tgttcccaag	attcaacgca	780
agctgagaaa	gaaaacagaa	acacgaggca	taaggccagt	cagaaaatgc	attacgaggg	840
gagttgaaat	gactgtgggc	cagtgccctg	gtcgtcatgt	gcaaattttt	tcttcgggtt	900
gactgccgca	gcattgcagca	acccgtcccg	ccctatataa	ctagtgtctc	atcttcttgt	960
cattcttctc	tctttaccaa	cacaccggtg	ctctttcacc			1000

SEQ ID NO: 56 moltype = DNA length = 1007
 FEATURE Location/Qualifiers
 source 1..1007
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 56

cttgacgggt	tcaatattca	atactcagtg	ttcaagcagt	ggtaggttgc	tcctgggatg	60
gaacgacggg	tttaatttct	tccgcacatg	accatgggtc	atgatcggtt	cagatggaca	120
gttggataga	gaaagtggta	acagataaat	agaatttctg	acaagtgtgc	aagagcctgg	180
ccgatggctc	atggctcgat	ggctcgatgg	ctcgatggct	cgatggccac	acaccatgtc	240
atctacttct	tgtgcttcta	tgcggacgag	tcggaacagc	tgtcaaactg	cgtttggggg	300
atctgagcgg	aatttgaaac	ctgtgtgaaa	ctacattgtc	tcctcgttac	ttctctggca	360
tctgtttgat	cgaaacggct	ctgcaaatag	acacgcgatg	aggcccaacg	acgggtgacgt	420
tgaacgcctg	gcccacctgc	tacaacaccg	tagaacgtgt	tgaagttgaa	cgtcggacgt	480
tggacgttga	gttgacgcgt	gatgctggaa	ccgtggatgt	agcagcgaca	gacaagccct	540
ctaagcaatc	tatgcatttt	tcaaaccaac	taaacgagcc	ttagttcgga	caaccctagt	600
ccagtgcggg	ggcagcgtgg	ggtcagttgc	cagcgaaagt	cacccagct	agctgggttg	660
gctagcttga	ctggctcaga	agtatcgatg	ctggagcttc	aggccagtag	gaaaatacac	720

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tgtcgtagtg ctccttcceg agttctgggc ttctgagccc ggccggttggt catcggtgca 780
cagtgccacgg accggcgggt gggacaacaa actgtctggg cgtgaaatcg tgcgagtggt 840
gatgcaaacc ggacgtaccc agcacgcagt tcccagtact cagacatact cagtacctca 900
gtgggtgagc ctcagcatcg cctccgtatt tctacaaagc tgttgagtcg ccttttgact 960
ttttactacc accatccctt cctcttctaa accccttaca ttccaaa 1007

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SEQ ID NO: 57      moltype = DNA length = 1000
FEATURE           Location/Qualifiers
source            1..1000
                  mol_type = other DNA
                  organism = Moniliella pollinis

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SEQUENCE: 57
tcgacgacag ttagtatagt caactatgct atctagtagc caaagtaagt ctacattccg 60
ggctttgcag gggtttageta ttccattaga caatacttac aaccttcgaa gggcttcccc 120
agaattcaat tctatatattg ttcaatatcc acccatcgga cccgctctcc gtacccccctg 180
ttcgcttcgc cactacaacc atgtcaagac ctgagaatcg tagcaaatg cccactacg 240
ggccgagtg cgtcggtggg ttaaagtaga atgacttggg tagccagatc gagacgctg 300
acattgatac gcagacagcg cagtcagttt caaactgaca actgacgcgt actggattga 360
caggactcct tggccgggtgc cattcgaatc gctttagagc ctcttgccag catcttgcca 420
gagaagcttg aaggctggat cgctcagcct gagctcacac ttagtagctc cactagtcc 480
tcagggtgct agcctctcct cagcgccctca agctcaagcc tcagggcctc agctgcatcg 540
agctctgggt ccagtgccag tgccagtggc agtgccagtg ccagtggcag tgccagtga 600
gagagtggtg tgagccaagc tataccacaa agcaaatgat gggtattgct agcttccatg 660
caccctcggg tgaaggaca acgaacaaca gacagagttt gaattattca ctggcctagt 720
actaccaatt tcaatttggt ggttcagcgc aggaacggcg cgagtgaagc taaggggcaa 780
ctccgcattg ggcgtgacgt cagtgctcct ctggctgagg cactccagca ggctgctgc 840
aacgcatcaa cgtgtgtgtg tgtctaccaa gcacattgtg cgctcgctgt ccgtgcgccc 900
aaatcgacac agtcagaacg tctataaaac cctcttgct tggagggtta ccgggcccgtc 960
ttttcccaac gctccgctc actcactact ctacacaaca 1000

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SEQ ID NO: 58      moltype = DNA length = 997
FEATURE           Location/Qualifiers
source            1..997
                  mol_type = other DNA
                  organism = Moniliella pollinis

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SEQUENCE: 58
tcggacccca agctctctag ctccaagctt tcaagctagc ttccagccag ggtcatggaa 60
ccgtaggagc tgggaaacgt gacgtgtgtg ctatgcctat cccatataag aattaacgtc 120
gaatcatgag gcgcatggga ggtgtcattg gtggatgatg acataacgtt atgattctat 180
atgagcagat gagcgcgccg gtctgcccgt agtgtggaag ttcgatacgc ccagtccatt 240
tcgttcagat gacgtttggt ccagaagtag ggtacatacc gagcgcaaac gtcgaactta 300
tggccttggt cgctcgtggt ccggcgctcg cgacacttcc atcaacggac gaaaggtaga 360
gccgaacgcc agcgcccaat ggcgatggcg atagtagaat gaatagtcgt cagatctagg 420
tcttaggcta caggccaccc tgccctgagg ctgggggtct tacagcgagt ctaaaaaaacg 480
aaaagccaac gtgccactt gggcctgccc atgcaccctg ccagagagcc ccaagctggt 540
ctgatgcaag cggagcaaaa tgttggaaga atagtttcaa cgatgggttg gttccagtac 600
gcacagtcca gagatgaggt actgcgcaac gagttgttgg cgacgggaac gaagatggat 660
tgaactattt gatctacagg tgatggcgtg gactcgggcg tggggccgct gtgtgatag 720
tagaagtggc gctattcgag ggtcgcgccc gtctggatgg atgcagaatg aagcttgacg 780
gccgagctg caggatgggg ccgagaaccc ggtgaaacat agttaaagcg cggggccagt 840
gtcatatttc gcttgagcca ttctactcgt tggcagccca ggccgcccga atttacgcca 900
actcttggtg ctcgagagct cgggcaatgt ggtatataag cttgaggttc cttacagcaa 960
agcctggctt cttaccacac accgcactct agcaaac 997

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SEQ ID NO: 59      moltype = DNA length = 1002
FEATURE           Location/Qualifiers
source            1..1002
                  mol_type = other DNA
                  organism = Moniliella pollinis

```

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SEQUENCE: 59
gagcgtagag agctccatcg atggcctggt tacaagttgc ctgaagacca ccaatacggc 60
ccgcacgcca acggcaggcc aagaaaggat aattgacaac cgtgaatctg aaagcttcgg 120
cgctcaggag ctcatgtgtt ttgttgtaga tataaagggt tctgaatacc agttgggtat 180
tgacacgggg agttggactg aatctctcag atctcccaga tctcaagtac ttagtcccaa 240
gcgtgccata attaaactta ggttgcgccg gacacggcac ctgagatcag ggtgtacgca 300
catactaact cattccacca ccaaagcaag actgagggta ccgcctgagg ttctggtat 360
tccaacagtc acaagtcaac gcgattgcgg tcaaatccca aagcgtagta ggcgctcgac 420
ggagggatga tggccagtag ccaatttcga gcactcagtt agatctgttc gtgtttagtt 480
gctcctgcag atctatctta aatgtatact gtgccaggta ggatctctcg gcacctatag 540
atgatataaa agatatgatg atatgccaa gtcgaagccg tcaaaactgac aaaatgaaag 600
aaacagcagg caagctaggg aagcgtctat cggggggatta aacggatgag tcagcagcac 660
aggagcagtc agtcgacaat tccggctttt tcccgcgccg aaggcccccag accccagacc 720
ccagccgcca ccattgctcg atccgcggg gttagcatga ccacggcaga tttctccgct 780
cagcttgctc ggacgactat atataaagaa tctttctctc gcagtcgctc cgagcccttc 840
ctccacccca ctggttacca ccaacgacat cacttggtct tgacgcccgc ctgcgttttc 900
ccgcttcgcg gtctccctag tgatttagct ggtgttttcc tctaccgcc actctactgt 960

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ttcatcgctt gttgccaagt tttctcagtt cattctcacc aa 1002

SEQ ID NO: 60 moltype = DNA length = 1002
 FEATURE Location/Qualifiers
 source 1..1002
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 60

tgacgctgga	caagcaccct	gaagtcgctc	agggcttcca	gatcaaggct	ttgcctaccg	60
tctacggctt	gtatgagggc	aagcctgttc	gtcagtgtaa	gttttaaaat	atattgttct	120
tgtctgttgc	ggttttacat	cagtgggtga	ctgacacggc	tttttgtttt	attgctgcag	180
tccaaggtgt	acttctctgag	gatatcctgc	gccaatattgt	ccaagatctt	cccagcttgg	240
ccgcccaaga	tgccccagaac	cgccctgctg	aaagtagcga	ggctgccaac	taagatatgt	300
ccatcctatg	agctgtccta	ccccaacaa	ctttttcacg	cccaacgtcc	aaacgtcact	360
tcttattgcg	gcaagaacaa	gctgcttccc	cactcattca	ttaccccttt	ctgatccacg	420
agggtgacct	atggcctttg	tacattccat	gaaacaggta	tccttttccc	caccattaca	480
tctgaactta	caaagacttt	gcttctctct	gtacacagct	cgtcgtccat	cgtccatcat	540
cgatgacgat	catcctgcat	tccgtatcat	gcacatcat	gcagcgatgg	caacgagggc	600
catgcaccat	gctgcataata	ctatgtgtat	agccaatagc	caatcttcaa	tctccaactt	660
cctacagaca	cagcctgtgg	ggaatcgctt	aatcgctgtc	catgcgtcca	tgctccgctg	720
tcctcgcgct	cgtcgcgccc	tcgcgacgat	taagtcacag	tcacgcccac	acagcgcgcg	780
agcgccagag	ccaccatcat	cacgtgatga	gacaaagagt	ccacttccgt	aaaagcaatc	840
ttaataatac	atgtaacctt	agcaatgtag	ccctaataat	ttgtttggcc	tttattttat	900
tgcgcccccg	cagcttttgc	accccgagga	gtaaaagtat	tcgggttgct	cgtcagagca	960
tccttttctt	tctccatcag	caaagcgtgc	gaccgccaag	aa		1002

SEQ ID NO: 61 moltype = DNA length = 1002
 FEATURE Location/Qualifiers
 source 1..1002
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 61

tttagctacg	atgggtgtgcg	tgtaactgat	cagtacaccc	gctgtatcaa	aactctatct	60
tcaatccgct	aagaatcgct	accatcagta	catcagtaca	tcagtgggct	aagatctaata	120
gggtgctgtc	acgggcccgcg	ggcactagcc	acgagccaag	agcggtagacg	ttgctgcgaa	180
atgggtggga	gcctgacagc	ttcagtttga	ataggatgaa	aacgccatga	agaacagaga	240
agctggcaat	tgacgatagt	ggggcgagcg	cacgctgctt	gctgggtccga	gcgtaggacg	300
aacgacaaac	cattgggccc	gccaccagcg	cccaggagg	gcgtcactgg	ttggatgtcg	360
tctgttaggc	tagagctcag	gcgctcaggg	cctctggccc	aagcctcgga	cgcaagcctc	420
tgacgagcga	gcgcgcgctg	gggcgggtct	gcgcaggaac	ggcaggactg	ccgagttggt	480
tgaatattga	atgggtggat	gattggtttg	ctgggttttg	ttttgacacg	caaggcacag	540
gacaagttag	tttgagggtt	ggatttgccg	aggtagaaag	ggcgctcgcc	ctcagggcct	600
cagcgtcagc	agacgagctc	gccagtcag	gtcagtagcc	agtgttcgtt	acttcccacc	660
gtccagagtt	tgctgctgcc	tctcagcttg	attcgtgtcg	gagaagtga	aatgggtcaca	720
cggtaccaca	cactgtcccc	atgcctgcgt	cctgtctcac	gagtggttag	ccacaaaaat	780
ttatggacaa	ccttactgag	cctgtgctgg	gggggtgctt	agcaccagca	ctggcgccat	840
ggcgtcaagg	ttaattccac	tgctcattcg	caaccttcaa	ctgatggggg	caactgggca	900
caaaaccttg	gatatataac	caactccaac	tcccacggtt	ttttaccttt	tgctctttct	960
cccaggactc	actaactcta	ctcctctatt	catcttcacc	aa		1002

SEQ ID NO: 62 moltype = DNA length = 921
 FEATURE Location/Qualifiers
 source 1..921
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 62

tttgatgag	gcattcaatg	tctccatgtt	tgaagaccag	agaaagtgtg	ggtaacacaa	60
tagccatgga	caattagcga	atccgaatat	gtgtactaac	attgtgcctc	caactgacag	120
ctggctgaat	aatgatgtga	atcatcaaac	atatcagaga	tatatccagc	aaccatttaa	180
caactgttat	tgtatgtatg	cattactgtt	ttacattgtg	tctttccaaa	tactaacatt	240
gccattatcc	aatccatata	tatatatata	tatagattac	acaacgcaaa	cctcgacagt	300
cgattgggtt	acatcgtgta	agcagccttt	tacggaatat	gatggacaac	cagatcctca	360
tagatggaga	caatgtgcag	atagattcca	tttatgatga	cattgatgtg	tttttaaatg	420
caaaataatc	tccatatata	gtgagttccg	agtctcccca	tgcttatctt	atcgtgtttt	480
ttctgtaaat	taggtcaaa	tattcaaa	attccctcaa	tctttcccat	ttttgtctga	540
gcccccatat	catcaaat	acgtatcaaa	tttcctaatt	agtagccccc	tatgttaaat	600
tagacatgtg	tgactgccat	ccgggacagc	ctacccaatg	acaccgcgcg	accgcacata	660
ttgtgtttag	tgccgcgcgt	ctgctgaagc	gactccctgt	ttgggaggaa	ccgagggcgg	720
gttggccgat	cccttgcccc	tcgctcctcc	tcctgggctc	cccttgccag	agggacaccg	780
aggggatccc	tcgtgtgaga	gcttgagggt	ggatgggtgt	caattttctc	atttgattga	840
aagagttgtt	atattgaaaa	ttcagtttgt	ggaagttgtg	attaaaaggt	tttactgttt	900
gttctgtaga	cacattcaat	a				921

SEQ ID NO: 63 moltype = DNA length = 500
 FEATURE Location/Qualifiers
 source 1..500

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mol_type = other DNA
organism = Moniliella pollinis

SEQUENCE: 63
acggatcaac aaaaaatgtgc gccagctttg ttgtgtgccc ttttctcta gcggatccta 60
tccacatttc gtgaatgaac cgattaagct ctgacgagat ttcttgttt tctgtagata 120
cctctttag aggtgtcaaa gttatgttgt gatcgagaga gttacctgca aggtattatg 180
caagcggtt gatgaggaca ctcaactacg tgtgagtagg agtgcggtgc tcagcagttg 240
acggtgagc cgcttgagca gcttgataaa ggtcctaata gctttccagc tgttggcaga 300
cgacaatgtg ctgtagtagt aggataggcc cgtaaaagtg tagaagcaat aggttgagac 360
tgactgtaat gggactttgc gttcaaagac tgaaagtcca aaggattgct caaaggcccc 420
tgaggtctga gcctctgagc gcgctcttcg tcaaaattcg ccggggccat cggctcatct 480
ctcgagggca gcactcttct 500

SEQ ID NO: 64      moltype = DNA length = 500
FEATURE           Location/Qualifiers
source            1..500
                  mol_type = other DNA
                  organism = Moniliella pollinis

SEQUENCE: 64
gtactaacca agtctattgt aaagctgggtg tctgctgctg cttcagcaag tgccattgct 60
accatcgcaa gcaaatgca aaaaaattcg ttcaatggtc taatttaagt aaaatactca 120
taatttataa aaagacgctt gtccttacct tttgtcttc tcctattat agtagctgta 180
gttatcgtag tctcaggac ttctgcaacg ctgcttgatg atcatatata cttcacctca 240
ttcattcata aagcattcat taggattgta tgatccggca gtgcatcatg catcatgcat 300
catgcatcag gacatcatgc atcatgccgt atcggtcgat cattgcattt gattggataa 360
tacgatctg tttaggcatt ggcagtcgaa acgaggataa agaagaggtc agtaccagtg 420
cacagcacgc gctcgcgctc tatgcgcaact ttgacgcttg aaaaaaactt ttcagccaat 480
cgatataga cagcggacc 500

SEQ ID NO: 65      moltype = DNA length = 493
FEATURE           Location/Qualifiers
source            1..493
                  mol_type = other DNA
                  organism = Moniliella pollinis

SEQUENCE: 65
atggctctgag ccgcccacga ttctcttctt catgagcgta tgacgacctt ttacttagc 60
atgtgctctc aaagaatacg ttttgcatac ccacatatc gtaaccctgt cctctatagc 120
ttctacaggt catagtcact ggctgccagt ataaacgtgt caatatctaa tctcataag 180
atgctctcgt ctgtcgtcca caataaaatt cagtaacgca aggactgcgc gactgagtga 240
ttgccgatga tatgagaaaa ctaatttcaa gccatttgaa cttcaataaa tgcttcgct 300
aacatcacgg atcatcgatc atcaaccagc ctccattaat aggcctgcca ccactcgcgc 360
aatcatgaat ctgatcatac ctggctagcg attctcataa caccagaatt acctgccat 420
taaagaaacc ctattctgcc tgtaaccccg caccagccgg tccgaattca cgacagacta 480
aggctctcgc gca 493

SEQ ID NO: 66      moltype = DNA length = 500
FEATURE           Location/Qualifiers
source            1..500
                  mol_type = other DNA
                  organism = Moniliella pollinis

SEQUENCE: 66
atcagttgga atcatcaacc tttttttttt gtatcgattt cttcctaccc tcacctccct 60
tttccccctt aactcctcgc tttgaacctc cttacccgag tgctacacct taccttttcc 120
ttttctcttc tcccctgcct aatattctag gcttgctact atactgctct ctctaacacg 180
atttttgcgc ttgtaagcat ttttctcttt ttaaactctag gttttatcat ttgttgcat 240
tatccaagtt tttttttttt ttgtttgta gtggttttac tgtttttggt tttttagct 300
tgactaggtt cttattctat tgtagatgta tatgatcgga tgcattcccc atctaaaagg 360
aaaatgtttc cgactgatat ttatttttac cttattgtct tcctaaatgg ttgactctt 420
gcctctcttc ttgggacgct ttgcaacggt ttgtacgtac gtaaggggtc atcgtttgga 480
taagcgacac agttgtcaag 500

SEQ ID NO: 67      moltype = DNA length = 324
FEATURE           Location/Qualifiers
source            1..324
                  mol_type = other DNA
                  organism = Moniliella pollinis

SEQUENCE: 67
attcacccgtg gagtaactgc gcttcacgtt ttctttgtta cgacaataat accacgctgc 60
atgctttgaa ttgagtatat attttgatga ctggttcaaa tagatcttcc ctccatagcc 120
ttcgagtact cgtgatctca gcattatcag atttgtgtcc agggccccga ggacgatgcg 180
gttttctcgg agtcactagt tgggaagatt tgtactggtt ggggtgtaggt agagagaacc 240
attgattttg gtgtagctgt tcgttttata agggcgagat tacgtaatgc acatgctgcc 300
ctagcaaatc gcgtcgctgc acgc 324

SEQ ID NO: 68      moltype = DNA length = 496
FEATURE           Location/Qualifiers

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source                1..496
                      mol_type = other DNA
                      organism = Moniliella pollinis

SEQUENCE: 68
atgtgcttag attaacatac acttgacttc cttacgcgtt gatataact tggatctcca 60
taacttctat tgtatagttt aatcaatgga ctccatgttg acatoccatg atgcgatcta 120
tagtagctgg aaatagcaaa attgaagttc ggcagacgac tgcaccagga ggggtagatt 180
ctgtcccata actcctacc acaggccgtg ctctgggtcg attacctta gctgatgctg 240
ctcaatgtaa cgcgtaattc tgagtgggtt tcggctgctc aagggttcag tatgcgctgt 300
gcagatagta ctgtatcgtg aaactcgacc gttgcacatg cactttatca agtgcgtag 360
gcccagagtc gtaggccaag agtcgtaggc caagatggaa ggctgaaga aaaagaaagc 420
tttgatcgtt ctgatctatg caatggattc atgagctaag tattagctat acaaatgatt 480
cgagcatagc agaagg                      496

SEQ ID NO: 69          moltype = DNA length = 500
FEATURE               Location/Qualifiers
source                1..500
                      mol_type = other DNA
                      organism = Moniliella pollinis

SEQUENCE: 69
gttgcctcct ttgctgttct atgatccaaa agatgcttag tgctccactc tctactcccg 60
actacctttt acgcattctg acatgactag tagtgcaatg ttacctctgg tgtttctctg 120
caatcacttt ctgtctgcta tagcttttga tcgtcagtag ttttttggtg tagctaaagt 180
agacttttgt acctggcggt caatgtaatt ttactccttt tttgtgcgat gcaacagatg 240
gtgtaaaagt cgtcgtaaaa gttttggctt gatgcttgat tcggtgacgg ttgcttgggt 300
gatccttata tcgtctcctg gtgttctcat aagcccaggc tcttcttagc tatgtacaac 360
aggccacca ctgactatgt tatgtagagg catgcattta acttacggaa agatttaagt 420
gagaagaacg tacctaagtca gatgcttgac atgaaaacgt attcctgcaa ttcacaaatc 480
agtatcctgc ctgtccactc                      500

SEQ ID NO: 70          moltype = DNA length = 500
FEATURE               Location/Qualifiers
source                1..500
                      mol_type = other DNA
                      organism = Moniliella pollinis

SEQUENCE: 70
gcgggttcgca cttttatccc aggcgatgaaa aatgacgcac ggctgtcgca cgctacaccc 60
agtagtcgcc ttttcaggca gtctatgagg ggagcttttc attgcccata attactagcc 120
tgcaattcat gatatacagg tcactttctc ccgatgtcta tagatttctc atagcgcttt 180
gaaaagacaa tttagctact cagtttataa gcaaatgatg tggaggtatt tattaggtac 240
aagatacgat aacgtttaca cgcggggaag caagtggcaa atcgctgaca acaccgcagt 300
gtggtggtgg aatcgtaact catcaaagct tgctgtgcac tttgggaggc accaggttga 360
ggccatcgcc cgaagggtgtt tgtggccagg gtttcgatat ttcataaatc aacgtgttga 420
tggatttga ttgggtaaga ggtcgatccg gctgttggtc agagtgtcga tcacattcgt 480
cgttgtcaag cccgagttgc                      500

SEQ ID NO: 71          moltype = DNA length = 500
FEATURE               Location/Qualifiers
source                1..500
                      mol_type = other DNA
                      organism = Moniliella pollinis

SEQUENCE: 71
gtggcggttga ttctgtatctt attattgac tgtcctgttc cccacgttgc caagtgattg 60
ttcatgctat cctctctcca ttctgtttgt gttgcacagt cggtagttgt acgtagtagc 120
tcgtcacaaa atatgaagtg agccttgcaa tgtaggcatt cacaagctgc taggcaggct 180
atgatctaag ctaagtggca aaaattctct tgattcaatg ccaaaactata ctattgttcc 240
tcaggcaaaa gcagctatat aagttgcatt caacgcaacc gccatacaag caaattagag 300
agaaaacggga tataaggtgc tttatccctg aaggagtagc cagcgtctac ttggttgcta 360
cacaatacac caaacagtgt acttaatgag ccattggctg aactggagct cgcagaggtt 420
cgagaagggc gggtcatacc agctgccctc catggaagct ggggtgtcga tgatgaaaag 480
tgtcggaaacg tacgtagta                      500

SEQ ID NO: 72          moltype = DNA length = 500
FEATURE               Location/Qualifiers
source                1..500
                      mol_type = other DNA
                      organism = Moniliella pollinis

SEQUENCE: 72
acgaagggtt tcccactctg gtgacctgat ccctttttac gtttttttgt tttgtttttt 60
cttttggtca tgatcaactg gggagtgatt cctcttggtg atgtcgaaaa ttgaactgtt 120
attaccaatat tctgttgctt ttgtttcctg tttgtagttt actacttttt acagtagtgc 180
ttgttctctg cttgcttgcg ttgtcagggt ttcccttata gcttagtacc aatatctagc 240
cttaattttg cttattcttt ctaccgccc atgattgatt tcaatcatga ttaatgattt 300
atgcatgaaa actcaaaaa ctcaattgtc ttgtacaagg caatcacaaag ctggtcagcc 360
attctcagcg cctataagag ctttcagcca ggccggcccc aaaacgatct accctgtgga 420
cctggctagg ctaaaactttg gaaagtcagt caattatgaa cgtttgaagc tcaagcgctc 480

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ctaccctctc ctggagcagt 500

SEQ ID NO: 73 moltype = DNA length = 500
 FEATURE Location/Qualifiers
 source 1..500
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 73
 tgggtgttcac cactgttgct tcccccttgt tggagcgctc gtcgttctga tggcgctctg 60
 agcccaaaat atcccttctt cgcgcgagag caagcctcgg agtgtgcttg caatctactt 120
 ttctactaa agttatcttc ttgtggcttc tgaatttgaa tagaacgtgt tcagacttgt 180
 tagctaaatt tgttgatgcg agatacttga cagaatcccg ttgatgaact catctgtgtg 240
 accaccgctt ggtgatggaa tctcgcgcgg aagccgtggt gttactgact ctcaagcaat 300
 tcagtagata aatttgata acatcattgc aaggggtgct agcaatcaat tgtccaatac 360
 caaacggcgg caagtgcata ggggcgatag agagtctctg cgaggaaggg aaggaatccg 420
 tatgcctgaa aacttttgta caagtgtctt tgtaaggcgg aggtctgtct tgacatctct 480
 ctgagatctg agatctgata 500

SEQ ID NO: 74 moltype = DNA length = 500
 FEATURE Location/Qualifiers
 source 1..500
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 74
 gtttcttggc tgttgacgaa atcttatgaa atgcacgggt ctaatatctc ctgccaattg 60
 tgtacccatc tctgttttga atgtcatagc tagctctgta gtttctaaac gaccgtctta 120
 ggccgggggt tctgtattgg ggattaagga taggatactt ggccagtttg ggccgatcaa 180
 tatcaatctg ctaggcgctc cttagggtcaa cttgaattat gatatatgag gtatgaggaa 240
 agtgggctaa atctgtggat taggatcaga aaaaacgtgg cagaccctcg atctaggcct 300
 acgtcagcaa tttcagcaat ttacgcgcct tgcctcttag ataggtttca gccgtctcagc 360
 gcgcgcaatt tcagcatgga tgagatcaaa ccaaggggca gcagtggcg accctgtagt 420
 tccaatcgca tgcctagtgc ataacgcgca catcaacgtg tcagattctc agagatctca 480
 acaagccac tagaccctag 500

SEQ ID NO: 75 moltype = DNA length = 504
 FEATURE Location/Qualifiers
 source 1..504
 mol_type = genomic DNA
 organism = Moniliella pollinis

SEQUENCE: 75
 aatcgctctg gtatogaaac gggccagctg ggtaaggcca gcggccggta gctaggacta 60
 tgaggctacc aggtctgttg gtaactaggc agctaggcac ctaggcaact gagcaactga 120
 gcaactgggc aactgagcgg tataggtttg gcttgggcac agttgatatt gatatcaacc 180
 agtcgctcgt tctgtttcac agaaccagct gtgtggtagc gagcgtagaa tagccaaaag 240
 gttttgagct gaggcgctga ggcgcacagg gaggacatca ccgcctgagc acacaggatt 300
 tcaaaactgtc agacggcgct ggtgccagtg ttggaagtgg gcgaagcacc tcgcattctaa 360
 gacatgtaat taaactctcg acagccatcg tccattgaca ccaacgtgca ccaacacacg 420
 cgagagctgc ttcgttgtgc ctgtcgacga cattgcctcg ttccccagcc aagtcgcccc 480
 accgctacac caagcgcact cgca 504

SEQ ID NO: 76 moltype = DNA length = 493
 FEATURE Location/Qualifiers
 source 1..493
 mol_type = genomic DNA
 organism = Moniliella pollinis

SEQUENCE: 76
 cctttgcctc acgcctctcg tttgcacgct tcttcacggc ttatcacgcg ttatttatc 60
 catggaatcc cctccaaagt tttcttcaac atagcttcat atcttctacc ataactctta 120
 tcccaccaag tagcacatga attcaacgta gcttgagaac ttttctcgt cagctgattg 180
 cgttctttag gcagttttgt tctaccccaa ttgctttcac cccaattgct tctgcttatt 240
 tcaccagtcg gctcggagat ccttgacaac tctgagatct gatgatcaac ctcttgtgca 300
 acttgtaaaa tttgatcgat gagacgatta gggacttgaa gctttaagtt agacgcgatg 360
 ggccggtaat tatatgctta gactttgttg ctatgtttgt cctataagat ctgatgcattg 420
 atgtatcatt accagacagg gctatcgca aaggtgcaga tgcatcttat catgacagtg 480
 ccacactggt ggc 493

SEQ ID NO: 77 moltype = DNA length = 499
 FEATURE Location/Qualifiers
 source 1..499
 mol_type = genomic DNA
 organism = Moniliella pollinis

SEQUENCE: 77
 catccgaaat tgggtggttg acaaatgcac cttcttctcc aggggctcgt gcaaacgcag 60
 tagccgagaa gaggcgctacc ggagaggaag acccaacaac cactgatcgg tttgaagttg 120
 gatgagaaag ttgtgccaat cgcggagccg tagcaacgct acgagcgcta cctacaacga 180
 ctccacgcgc aatcggtacg cgcaaggaa cggatgagcg accgagcagg ttgcgcgaag 240

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ccatgacgat tgcctctacc ccaacgagct ggatcaagtt gggcagaaca taatttatct 300
cgcttccccg cgccacaagt agctccacca ttcttctcaa aatagccoga ttccgctttc 360
ggggggcaaa acttccgcct ataccgatcc gattgaaacg ccaaaaaaat taagttgtag 420
tacgcctgtt ggtcgagccc gactggagca ccatttgaga cactaggaga taatttatct 480
ttttccttct tggcccca

```

```

SEQ ID NO: 78      moltype = DNA length = 501
FEATURE           Location/Qualifiers
source            1..501
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

```

```

SEQUENCE: 78
cctgaacgtc ctgaactctg tcttattaca taccattaat ttccctgcct tccctctcta 60
cgtcatcgga attttatctc ttcttgagca gcaaagattt ggtccttgtc cgggatacaa 120
cggagagtca tgattgggtt tatggaagag aggcgatagc acttgcatth tgaccattcg 180
gggattaagc gaacccccaa gtggcacaaa gagctcgtat gagggtcagt aagtatagag 240
ggacatgcaa aggagagaaa aggaacacaa atattgacaa atagagacaa gtacaggtga 300
tcgcaatagc atgatgtagg ggtatgcgca agtcacaatc gttcaggggc ggcataacat 360
cagcgtaccc gagcacaatg gttaatgatt aatgagcaga ggaaaaaata atgattatgc 420
cggggatggc aggaactaact ggcgagctc ttcaagcttcg gcaacgcgtg cacatagtg 480
tgtagacca cgagattgag c

```

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SEQ ID NO: 79      moltype = DNA length = 300
FEATURE           Location/Qualifiers
source            1..300
                  mol_type = other DNA
                  organism = Moniliella pollinis

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SEQUENCE: 79
cgggtaagac caagtcggtt ggtgtttcga acttcagcac tcgtctggtc gacttggttg 60
aggaagcttc gggcgaacgt cctgcggtta accagatcga agtcacccc ttgttgcaac 120
aagacgagtt ggttgctcac cataagagca agaacattgt catcactgct tacagtcct 180
tgggcaacaa tgcgtctggt aaaccacctc tgactgagaa ccccggtatt gtggtgctg 240
ctaagcgttt gaaccatact cctgctgctg tgctcattgc ttggggtatt caacgcgggt 300

```

```

SEQ ID NO: 80      moltype = DNA length = 400
FEATURE           Location/Qualifiers
source            1..400
                  mol_type = other DNA
                  organism = Moniliella pollinis

```

```

SEQUENCE: 80
aaccagatcg aagctcacc cttgttgcaa caagacgagt tgggtgctca ccataagagc 60
aagaacattg tcatcactgc ttacagtcct ttgggcaaca atgctgctgg taaaccacct 120
ctgactgaga acccgggtat gttggatgct gctaagcggt tgaaccatac tcctgctgct 180
gtgctcattg cttgggttat tcaacgcggg tacagcgtct tggtaagtc agttacacc 240
tcacggatca agagtaactt tgaacagatc actctgctg atgaggaatt ccaacgggtt 300
accaacctca tcaaggagta cggtagagag cgtaacaacg ttcttttcaa ttacaagcct 360
tcgtggtcca ttgacgtctt tggtaaccag gacgaggcta

```

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SEQ ID NO: 81      moltype = DNA length = 453
FEATURE           Location/Qualifiers
source            1..453
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

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SEQUENCE: 81
gcgaggagta gttactaggt agcggtatag caattccgcc cataatgcatt aaccgttctc 60
cctacaaaaa caaaacaatg gctacgtggc cggcgctgct ggccgacatg gtgcgggtgc 120
ggggggcggg gtacgagtggt ggggagttca ctagtctgct ggctagctga ctagctgacc 180
ggcgtgtggt tgtggcgatg cgcagtggga aagattaaat atgaaatatt gaatatcaaa 240
atgacgtttg ggtgggcacg tgcgcgcgcc aacaacaaca agagcagcca ccaagccaaa 300
gcaccaagca ccaacccgcg acaccatttt gttacacccg cggcagaaat ggcgtcagcc 360
agcagtcact attactcaat attcaatagg gtttccactt gactcaattc ctcatgtata 420
tatatatcca tctaattctt ccaatacata acc

```

```

SEQ ID NO: 82      moltype = DNA length = 567
FEATURE           Location/Qualifiers
source            1..567
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

```

```

SEQUENCE: 82
gcttgctggt cacattattg tcagtacctt ccatgaagaa ttcaatatta ttttacattg 60
tcaaccatta catggtatccc atgttgtgca tagcctctat gtcacccctt ggtcttttat 120
gtctgtagcg tgctagtagt attcaagttg tagcactgct gtgaaacagt acgttcaagt 180
agtgtgcagc agtacgttca gttagtggac tggggcagcc tcgccttcag caaaggttag 240
gtacatgagc cagttgctca ctcttgcttt ctacgaatcc cataggctac ttgccccct 300
aacgtgagc accccctgtg ttgagaagt ttgagctttt ttacgttacc gtgaccaagc 360
gagcagagta gaaactcttg aagttgtgta tgtaagaata caaattgaga acacaagggt 420

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acctaccttt atagttagta aacagcaagc aataccagag attttcatcg caacgtgact 480
ctatcattca ctatattaag ttgcacacgg atgttgtaca acactacacg taaaagagag 540
aaaaatacta atgtcgatcg ctcaggt 567

```

```

SEQ ID NO: 83      moltype = DNA length = 500
FEATURE           Location/Qualifiers
source            1..500
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

```

```

SEQUENCE: 83
gcggcctatg cagacgcttg gagcttgggc acattggata ttggatattg gctggctgcc 60
gttcgagatc ccgttgctgt cgccaccgcc accggtcacg tggtcacttg gtcacgtgga 120
ccacctgcac tgcgcttagc tacaagagtg catgatgcac gatgtcatgc atgatatgca 180
tgatacaagc atgaatatgt atttcaatat tgaacattta gtattgatat ctcgatatgc 240
gtaaacacgtg acacgcacca tatcaatcag ccgcaccggc actaaatggt tcaatttcaa 300
catcccacac gctcgaatgc tccctcgagc taatcggatt tcggagctcc gacatggccg 360
gcctcgagcg gagtttgaaa ctccgttccg cccgaccccg ctggctgcgc cgatgaaatc 420
ccgagtcaag tcatagcaag ccaagtggct ggctgactca aaagctaata aaagtcgtag 480
caagtaatac tccgcacgct 500

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```

SEQ ID NO: 84      moltype = DNA length = 500
FEATURE           Location/Qualifiers
source            1..500
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

```

```

SEQUENCE: 84
aaatcttcat ccagcgtagc aagcatatcg cgcattgtccg agttcctcga gtgtgcatac 60
cgaaacccaaa aagccatgga tgtcattgac ctgctgtatt catcaatcga gatattctta 120
ctccggagcg tacagtggca gatattatgt aggagatgat gggacaggat gcagagatac 180
ataatccatt tccagttaat agggccgcta gatacttgat attagcctgg cagcgagcga 240
gcgagggtag agtagaatag agtccagtat agatcccacc ccatggtgcg ggaagtggat 300
gacgtaatag cgcgatagct ccccgctccc ttgtcaccta ggccacggga ccctacgtga 360
ttgcatcgtg cgtgtcgtcg cctgtcgacg cctggtgcag gcatacgtg cgtgtcttca 420
cgatacaaca caccaccaac caagactcca gttccggcca gccagtacca tggcgtcggt 480
ctttgggttt agtgcacgct 500

```

```

SEQ ID NO: 85      moltype = DNA length = 500
FEATURE           Location/Qualifiers
source            1..500
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

```

```

SEQUENCE: 85
tcgcgtcttc aaaccggtggc gcgcgacacg tgccgcgacc aaaatttggt ggtgtatggg 60
aataggggtt tctaaaattc tgttgaggat gtgagtatca ctggcatggc accagtgtac 120
tgctcatcga taatcataaa cgtatcaacc ctccggctct cgcattctga tctgctactg 180
gagggcacag caagccttct gcttggttgc gctctgcgct ctctgtggtg ctttgccgct 240
cgcggccaag gtcgcgacg gcgattcacc accatacggg ttaagccacc agacgagtgc 300
tgctcgccgg gccagctcgg cgcctgtcac gccccacat tggcttgatg ctcggtctct 360
tctttcatct tggaaccggc gggcaatggc gcttatcacc actcctgggc cgggccccgc 420
gtatactctc ataactagtg cagcgtttca aataaatacc tcgcctctct gctggccgac 480
cgacgtttac ccattccgcc 500

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```

SEQ ID NO: 86      moltype = DNA length = 500
FEATURE           Location/Qualifiers
source            1..500
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

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```

SEQUENCE: 86
atcctgtgct cgtctcttac attcatcttc ctgcccataa aactttcaag ctatttctc 60
caacgtactc aacaacctat ttttttgggc ccgctcaaa acgcacctta ttgcgtctaa 120
gtgtattcta accacgcgac ctgaattgca aagcctgtta ccccatcgc acctcgtct 180
taagatagtt gtttaacgag ccttggccag ttttagcctt gacctgact acagatcata 240
atggttagat gctcttgat cgataatgat atattgcag tcgacgattg aagatttatt 300
ggatgtcgat gacgactgtc gagctcttat tcttagcaac aacttaata ttcactatcg 360
aatgcacgag acatcatgca tcgtgctcat gaatatcact ggatcactct cgatcgacg 420
ccactaggtc agcataacac gcttgaagct gacttgaagc tttggaacaa atggcagcaa 480
cagtggcggt caggtgacc 500

```

```

SEQ ID NO: 87      moltype = DNA length = 500
FEATURE           Location/Qualifiers
source            1..500
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

```

```

SEQUENCE: 87
tgacagagct tgcaaatgat gaaagaccaa tgggaattgat cattgattat tgatgatgca 60
tgatgaaaga gatagaaaca gatgcaagat gggatggcgc gatgatcaaa tgataatgaa 120

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cgttcaaccc ttcaaaattg aaactctcag gctgagcggc tgagcaagct ctgagaatgt 180
gggtcttcga agactttcgc ttgagcatgg tgcattgtgca aagagatgga atgtggaatt 240
gttttaggta ggaacacagc ccggtagagc ggccgagtg ctcagcgaag ctgggagAAC 300
gactgaaccc gtctttagat tcatcatctc agcgtgcgtc agcgcgtcag ctccagcgtt 360
tgagaatggc acggactgac ctgtcgtaca tcccaaggag gcccggtgcg tatactcgcc 420
ttgtggccta atctcgcatc aacccatcct caacgaaagc gcaagtagcg actaagcaca 480
ctatctttcg ggtcgggtcg

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```

SEQ ID NO: 88      moltype = DNA length = 486
FEATURE
source            Location/Qualifiers
                  1..486
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

```

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SEQUENCE: 88
attgttgcac cgtctcatca gatcgtcctt cattgtagtg ataattgggt caaccttgta 60
ttagactgta tagcagaatc cctcgctata aaaagtatta tgcatacgca gttccaagca 120
ggaattagta tggtagtggg acttggtctga agggctgacg acatggcgag aaaaacgtat 180
caattcggaa cacactgtgc aatgtcgatc acgttttcgg tggttcacgg taggcttcct 240
agtccatggg aaagtggtaa catctatccc ttgagcgtt gagcttaagt ttacctgtg 300
caaaagtcac ttgaagcttc acttgaagct tgaagctttg aagttttgaa gcttgaagcg 360
caaacagcag cggtagcatgt atactggcgt aacctgccta cagtgggtga tcagtacgtc 420
aagcgggtag cccgaaatct cattaaggtc atttgccctc taagcttagt acctagtgt 480
ctgttc
486

```

```

SEQ ID NO: 89      moltype = DNA length = 499
FEATURE
source            Location/Qualifiers
                  1..499
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

```

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SEQUENCE: 89
tggcatgtag gccgttaggg gctatgccta tatgctcatg cactcatgtc gcgatgttgc 60
gatatcgcca tggaccaaac aagcaagcgt gcagcgtggg cgttggggc aagggtgcta 120
agctaagtgt aagcactgaa gtgctgaagt gctgaagtcg tgaacgtgga acgtcaggat 180
gacacatcct ctcccttcgg ctctgtccaa gaaagggttg tcagccatct gagatctgag 240
atctgaaatg aatataatg tgaattgagt cgaatgagtc tgactatctt gggagattgc 300
accaatgcgc cacggtgacc agatgtgtct ggtcttagct ggtatagtag cactgatct 360
gccttgacag atctcatgtt catacgttca tgcattcaat cttatcgtct agcgtctgc 420
gatctgccc aactagacgac gacaaatgta attgatcctt gtaaccctgt tgetgggtgc 480
agcagcgcca tacgatgcc
499

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```

SEQ ID NO: 90      moltype = DNA length = 504
FEATURE
source            Location/Qualifiers
                  1..504
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

```

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SEQUENCE: 90
ttgtgtgtac gacctcattg atcgatgac tccgacactt tgagttagctt tccatcctct 60
ctttactgta ctttgtatct ttgctacttt gtgtctcttc ttaattgttc cgaaccgggtg 120
attttgtgta ttagagtggt tgttattggt gtcggtactg acattgggtg ttccattaga 180
gttccagtag accctcatac cgttccagag tcccgcacac cacaaggatg tttcttagat 240
agggtgcagc aagatccaca aaaaagtgtg ggtatttagt ggtttcaagc aagggtacaat 300
attctaattc aatactcatt tgaataatca atattcaaca gcttctagcc tacaagcttc 360
aatgttctcc gtcgggttaac tcagtatact tagtactcag ttcaagtcag tgttggttaa 420
ataaaatcac gtgatccaaa aacttgaaac ttttcatctt tctcctcaga tgttggtgat 480
tcgtcccata gcgggttacca caac
504

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SEQ ID NO: 91      moltype = DNA length = 1679
FEATURE
source            Location/Qualifiers
                  1..1679
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 91
aagacctctg tcccacaac cttgatgggc tgtgaaaccc tcaacaacat cattggtaga 60
actgtcaatc caaagaacaa aaattgggtc tgtggtgggt cttctgggtg tgaaggtgct 120
attgttggtg ttagagtggt tgttattggt gtcggtactg acattgggtg ttccattaga 180
gtcccagctg ctttcaactt tttatacggg ttgagaccat ctcacggtag attgccatat 240
gctaaaatcg ctaactctat ggaagggtcaa gaaacgggtc actccgctgt tggctcctatc 300
actcactcgg tcgaagactt gagattgttc accaaatctg tcttgggtca agaactcttg 360
aagtagcact ctaagggtcat ccccatgcca tggagacaat ctgaatctga catcattgcc 420
tctaagatta agaattggtg tttgaacatt ggttattaca atttcagcgg taacgtcttg 480
ccacaccac caatttttac tgggtgtcga actacgggtg ccgctttggc caaggctgggt 540
cacacggta ctccatggac tccatacaag catgatttct gtcattgact gatttccac 600
atctatgctg ctgatggtt tgcgcagctc atgagagaca tttctgcctc tggtaggcca 660
gccatcccta acattaagga cttgttgaa ccaaatatta aggtctgtaa catgaacgaa 720
ttgtgggaca ctcatctaca aaagtggaa tatcaaatgg aatacttgga aaagtggcgt 780
gaagctgaag aaaaagctgg taagggaattg gacgctatta tcgctccaat tactcctacc 840

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gtcgcgtgtca gacacgatca attcagatac tacggttacg cctccgttat tagcttattg 900
gatttcacct ctggttgctg cccagtcact ttcgctgata agaatttga taagaagaac 960
gaatctttta aagctgtttc cgaattggat gctttgggtc aagaagaata cgaccagag 1020
gcttatcacg gtgctcctgt tgcgtgtcaa gttattggta gaagattgtc cgaagagaga 1080
actttgggta tcgcogaaga agtcggtaaa ttggtgggta acgtcgtcac tcataaatgg 1140
tgttcaccac tggtgcttcc cctttgttgg agcgtccgtc gttctgatgg cgctctgagc 1200
ccaaaatc ccttcttcgc ccgagagcaa gcctcggagt gtgcttgcaa tctactttc 1260
ctactaaaag tatcttcttg tggcttctga atttgaatag aacgtgttca gacttgttag 1320
ctaaatttgt tgatgcgaga tacttgacag aatcccggtg atgaactcat ctgtgtgacc 1380
accgcctggt gatggaatct cgcgcggaag ccgtggtgtt actgactctc aagcaattca 1440
gtagataaat ttgtataaca tcattgcaag ggggtgctagc aatcaattgt ccaataccaa 1500
acggcgggcaa gtgcaaaggg gcgatagaga gtttctgcga ggaaggggag gaatccgtat 1560
gcctgaaaac tttgtacaa gtgtctttgt aaggcgagg tctgctctga catctctctg 1620
agatctgaga tctgatacct gcaggataac ttcgtataat gtatgctata cgaagtatt 1679

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SEQ ID NO: 92      moltype = DNA length = 1903
FEATURE           Location/Qualifiers
source            1..1903
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 92
ataacttcgt ataagtatg ctatacgaag ttatctctga ggagctcaga ccttgcatgt 60
tagccttgac cttgcctcta tgaccactaa ttgcacaaag attgcacggt accattgtca 120
tcttgccatg tatccaatgt ttctgtttg ccatagtacg tagtagatac atcgatcatc 180
gatgcagatg acgcagatgc cattggttcg ctttgcccta tgcatgcac atgcacatg 240
catcatgcgt catgcgtcac tatccatatt aatcattgcc ctctgacta ggccatgaca 300
cctgcagacc cttcccagg gatgcttctt gtaatagcca atatactaca caactttacc 360
gattacgacg tagatcgatc tcaactatgt ttgtcgaact tgcgtgcaca tcgaggggtg 420
ggaaaaaagc tgacgggaagc ggaagtgggt gaattgggta atttagttag attggcaggg 480
ttgttcccaa gattcaacgc aagctgagaa agaaaaacaga aacacgagggc ataagggccag 540
tcagaaaatg cattacgagg ggaagtgaag tgactgtggg ccagtgcctg ggtcgtcatg 600
tgcaaathtt ttcttcgggt agactgccc agcatgcagc aactctgccg gccctatata 660
actagtgcct catcttcttg tcattcttgt ctctttacca acacaccggg actctttcac 720
catgccacaa tcttgggaag aattggccgc cgacaaacgt gccgttttgg ctaaaaccat 780
tcttgacgaa tgggaaggttc aaactttgcc tgccgaagat tccgtttattg atttcccaaa 840
gaagtccggt attttgtctg aggtcgaatt gaagattacc gaagcctctg ctgctgattt 900
ggtttccaa gttgcccctg gtgagttgac ttctgttgaa gtcacttttg ctttttgtaa 960
gagagctgct attgtcaac aattaaccaa ctgtgctcac gaattcttcc cagatgctgc 1020
tttagctcaa gctagagaa tagatgaata ctacgctaag cataagcgtc cagttgggtcc 1080
attacacggt ttacacatct ctttaaagga ccaattgcgt gttagggtt acgaaacctc 1140
catgggttac atttctggt taacaaaata cgatgaaggt gattccgtct taaccaccat 1200
gttgagaaaa gctgggtgctg ttcttacgt taagacctct gtcccacaaa ccttgatggt 1260
ctgtgaaacc tctcaacaaca gattggtag aactgtcaat ccaagaaaca aaaattggtc 1320
ctgtgggtgt tcttctggtg gtgaagggtc tattgttggg attagaggtg gtgttatttg 1380
tgctggtagt gacattggtg gttccattag agtcccagct gctttcaact ttttatacgg 1440
tttaagacca tctcacggtg gattgccata tgctaaaatg gctaaactta tggaaaggta 1500
agaaaccgtt cactccgtcg ttggctctat cactcactcc gtccaagact tgagattgtt 1560
caccaaatct gtcttgggtc aagaaccttg gaagtacgac tctaaggta tccccatgcc 1620
atggagacaa tctgaactcg acatcattgc ctctaagatt aagaattggtg gtttgaacat 1680
tgggtattac aatttcgacg gtaacgtctt gccacaccca ccaattttac gtgggtgtcg 1740
aactaccgtt gccgcttttg ccaaggctgg tcacaccgtt actccatgga ctccatacaa 1800
gcatgatttc tgtcatgact tgatttccca catctatgct gctgatgggt ctgccagct 1860
catgagagac atttctgctt ctgggtgagcc agccatccct aac 1903

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SEQ ID NO: 93      moltype = DNA length = 876
FEATURE           Location/Qualifiers
source            1..876
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 93
agatgtagcc ggcgctgtag agttttggac ggataggttg ggtttcagtc gagacttcgt 60
tgaagatgac ttgtcaggcg tcgtgcgtga tgatgtgacc ctgtttatct cggccgttca 120
agatcaagtt gtcctcgaca atacacttgc atgggtctgt gtacgcggat tagatgaatt 180
gtacgcggaa tggcttgaag ttgtatccac taactttcgt gacgcttcag gtcctgctat 240
gactgagatt ggtgaacaac cctgggggag tgagttcgca ctccgagatc cagctggaaa 300
ttgtgttcat ttctgtagctg aggaacagga ttaatggtgt tcaccactgt tgcctccct 360
ttgttgagc gtcctgctt ctgatggcgc tctgagccca aaatatccct tcttcgccg 420
agagcaagcc tcggagtgtg cttgcaatct acttttccca ctaaggttat cttctgttg 480
cttctgaatt tgaatagaac gtgttcagac ttgttagcta aatttgttga tgcgagatac 540
ttgacagaa cccgttgtag aactcatctg tgtgaccacc ccctgggtgat ggaatctcgc 600
gcggaagccg tggtgttact gactctcaag caattcagta gataaatttg tataacatca 660
ttgcaagggg tggtagcaat caattgtcca atacaaaacg gcggcaagtg caaaggggag 720
atagagagtt tctgcgagga agggaaggaa tccgtatgcc tgaaaacttt tgtacaagt 780
tctttgtaag ggcgaggtct gctctgacat ctctctgaga tctgagatct gatacctgca 840
ggataacttc gtataatgta tgctatacga agtttat 876

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SEQ ID NO: 94      moltype = DNA length = 1087
FEATURE           Location/Qualifiers
source            1..1087
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 94
gaagactgag tccgtctcag ataacttcgt ataatgtatg ctatacgaag ttatcctgca 60
ggagctcaga ccttgcatgt tagccttgac cttgcctcta tgaccactaa ttgcgacaag 120
attgcacgtt accattgtca tcttgccatg tatccaatgt ttctgttttg ccatagtagc 180
tagtagatac atcgatcatc gatgcagatg acgcagatgg cattgggttcg ctttgcccta 240
tgcattgcac atgcattcatg catcatgcgt catgcgtcac tatccatatt aatcattgcc 300
ctctgtacta ggccatgaca cctgcagacc cttccccagg gatgcttctt gtaatagcca 360
atatactaca caactttacc gattacgata gagatcgatc tcaactatag ttgtcgaact 420
tcgatgcaca tgcaggggtgc ggaaaaaagc tgacggaagc ggaagtgggt gaattgggta 480
atttagttag attggcaggg ttgttcccaa gattcaacgc aagctgagaa agaaaacaga 540
aacacgagggc ataaggccag tcagaaaaatg cattacgagg ggagtgaag tgactgtggg 600
ccagtgcctg ggctgctatg tgcataatgt ttcttcggtt agactgcgcg agcatgcagc 660
aactctgccg gccctatata actagtgcct catcttcttg tcattcttgt ctctttacca 720
acacaccggt actctttcac catggcaaatg ttgacgagcg ccgtgccggt gctaaccgcg 780
agagatgtag ccggcgctgt agagtgtttg acggataggt tgggtttcag tcgagacttc 840
gttgaagatg actttgcagg cgtcgtgcgt gatgatgtga ccctgtttat ctggccggtt 900
caagatcaag ttgtccctga caatacactt gcatgggtct gggtacgcgg attagatgaa 960
ttgtacgcgg aatggctcga agttgtatcc actaaacttc gtgacgcttc aggtcctgct 1020
atgactgaga ttggtgaaca accctggggg cgtgagttcg cactccgaga tccagctgga 1080
aattgtg                                           1087

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SEQ ID NO: 95      moltype = DNA length = 1049
FEATURE           Location/Qualifiers
source            1..1049
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 95
attgaagccc ttgatggatc attcaccaca gacacagttt ttccgctgac agctactggt 60
gacggattta ctctccgtga agtgcctgta gatccgcat tgacgaaagt atttcagat 120
gatgagatg atgacgagag cgtgatggc gaagatgggt atcctgactc gcgcactttc 180
gtcgatacag gtgatgatgg tgacctggcc ggatttgttg ttgtgtccta ctgggggtgg 240
aaacgtagat tgacogtga agatatcgaa gtccgcgcag aacatcggtg tcattggcgtg 300
ggacgagcac tgatgggtct agctacggaa ttccgtcgag agcgtggggc gggtcactta 360
tggcttgaag ttaccaatgt aaatgctcct gctattcacg cttatcgccg aatgggtttc 420
accttgtgtg gccctgacac ggcattgtat gacggtagtg cttctgacgg ggagcaagcg 480
ctatacatgt ctatgccctg cccctaatgg tgttaccac tggtgcttcc cctttgtttg 540
agcgtccgtc gttctgatgg cgtcttgagc ccaaaaatcc ccttcttcgc ccgagagcaa 600
gcctcggagt gtgcttgcaa tctacttttc ctactaaagt tatcttcttg tggcttctga 660
atttgaatag aacgtgttca gacttggttag ctaaaattgt tgatgcgaga tacttgacag 720
aatcccggtt atgaactcat ctgtgtgacc accgcctggt gatggaatct ccgcgggaag 780
ccgtgggtgt actgactctc aagcaattca gtagataaat ttgtataaca tcattgcaag 840
gggtgctagc aatcaattgt ccaataccaa acggcgggcaa gtgcaagggg gcgatagaga 900
gtttctgcga ggaagggaa gaatccgtat gcctgaaaac ttttgtacaa gtgtctttgt 960
aagggcgagg tctgctctga catctctctg agatctgaga tctgatacct gcaggataac 1020
ttcgataaat gtatgctata cgaagttat                                           1049

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SEQ ID NO: 96      moltype = DNA length = 1257
FEATURE           Location/Qualifiers
source            1..1257
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 96
gaagactgag tccgtctcag ataacttcgt ataatgtatg ctatacgaag ttatcctgca 60
ggagctcaga ccttgcatgt tagccttgac cttgcctcta tgaccactaa ttgcgacaag 120
attgcacgtt accattgtca tcttgccatg tatccaatgt ttctgttttg ccatagtagc 180
tagtagatac atcgatcatc gatgcagatg acgcagatgg cattgggttcg ctttgcccta 240
tgcattgcac atgcattcatg catcatgcgt catgcgtcac tatccatatt aatcattgcc 300
ctctgtacta ggccatgaca cctgcagacc cttccccagg gatgcttctt gtaatagcca 360
atatactaca caactttacc gattacgata gagatcgatc tcaactatag ttgtcgaact 420
tcgatgcaca tgcaggggtgc ggaaaaaagc tgacggaagc ggaagtgggt gaattgggta 480
atttagttag attggcaggg ttgttcccaa gattcaacgc aagctgagaa agaaaacaga 540
aacacgagggc ataaggccag tcagaaaaatg cattacgagg ggagtgaag tgactgtggg 600
ccagtgcctg ggctgctatg tgcataatgt ttcttcggtt agactgcgcg agcatgcagc 660
aactctgccg gccctatata actagtgcct catcttcttg tcattcttgt ctctttacca 720
acacaccggt actctttcac catgggaacg actttgatg ataccgccta ccgttatagg 780
acatcagtag cgggcgatgc agaagcaatt gaagcccttg atggatcatt caccacagac 840
acagtttttc cggctgacagc tactgggtgac ggatttactc tccgtgaagt gcctgtagat 900
ccgcatttga cgaaagtatt tcagatgat gagagtgatg acgagagcga tgatggcgaa 960
tatgggtgac ctgactcgcg cactttcgtc gcatacgggt atgatgggtga cctggccgga 1020
ttgtttgttg tgtcactctc ggggtggaac cgtagattga ccgtcgaaga taccgaagtc 1080
gcccagaaac atcgtggtca tggcgtggga cgagcactga tgggtctagc tacggaattc 1140

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gctcgagagc gtggggcggtg tcacttatgg cttgaagtta ccaatgtaaa tgctcctgct 1200
attcacgctt atcgccgaat gggtttcacc ttgtgtggcc tcgacacggc attgtat 1257

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SEQ ID NO: 97      moltype = DNA length = 1199
FEATURE           Location/Qualifiers
source            1..1199
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 97
cttaaacacg gcaagggtag cgttgcaaac gacgtaaccg acgaaatggt ccgtctgaat 60
tggctaactg agttcaatgcc actccccacc atcaagcact tcacccgtac acctgacgat 120
gcgtggctgt tgactacggc catccccggg aagaccgctt ttcaggctct cgaggaatac 180
ctgactcggg gcgagaacat cgtggatgcc ttggctgtgt tttgcgccg ccttcaactc 240
attccggctc gcaactgccc ttccaacagc gaccgtgttt ttaggcttgc ccaggcacaa 300
tcacgcatga acaatgtgtc agtcgacgct agtgatttcg atgacgagcg caacggctgg 360
ccggtggaac aagtgtggaa agagatgcat aagctcctgc ccttttcccc agactctgtc 420
gtgacgcatg gtgatttctc gcttgacaac ctcatctttg acgagggcaa gctcattggc 480
tgcatctgac tcggccgggt tggcatcgcc gatcgctacc aagatcttgc gatcctctgg 540
aattgtctgg gcgagttctc cctagcctc cagaagcgct tgttcagaa atatgggatc 600
gataaacccg acatgaacaa gctccagttc catttgatgc tggatgaatt tttctaatgg 660
tgtttcaccac tttgtcttcc cctttgttgg agcgtccgtc gttctgatgg cgctctgagc 720
ccaaaatatc ccttcttcgc ccgagagcaa gcctcggagt gtgcttgcaa tctacttttc 780
ctactaaaag tatcttctga atttgaatag aacgtgttca gacttgttag 840
ctaaatttgt tgatgcgaga taactgacag aatcccggtg atgaactcat ctgtgtgacc 900
accgcctggt gatggaatc cgcgcggaag ccgtggtgtt actgactctc aagcaattca 960
gtagataaat gtgataaaca tcattgcaag ggggtgctagc aatcaattgt ccaataccaa 1020
acggcgggaa gtgcaaaagg gcgatagaga gttcttcgca ggaaggggaa gaatccgtat 1080
gcctgaaaac tttgtacaaa gtgtctttgt aaggcgaggg tctgctctga catctctctg 1140
agatctgaga tctgatacct ccaggataac ttcgataat gtatgctata cgaagttat 1199

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SEQ ID NO: 98      moltype = DNA length = 1475
FEATURE           Location/Qualifiers
source            1..1475
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 98
ataacttcgt ataagtatg ctatacgaag ttatcctgca ggagctcaga ctttgcattg 60
tagccttgac cttgcctcta tgaccactaa ttgcacaaag attgcacggt accattgtca 120
tcttgccatg tatccaatgt ttctgttttg ccatagtacg tagtagatac atcgatcatc 180
gatgcagatg acgcagatgg cattggttcg ctttgcctta tgcattccatc atgcattcatg 240
catcatgcat catcgctcac tatccatatt aatcattgcc ctctgtacta ggccatgaca 300
cctgcagacc cttccccagg gatgcttctt gtaatatgca atatactaca caactttacc 360
gattacgata gagatcgata tcactatagt ttgtcgaact tcgatgcaca tcgaggggtg 420
ggaaaaaagc tgacgggaagc ggaagtgggt gaattgggta atttagttag attggcaggg 480
ttgttcccaa gattcaacgc aagctgagaa agaaaaacaga aacacgagggc ataaggccag 540
tcagaaaatg cttactcagg ggaagtgaag tgactgtggg ccagtgcctg ggtcgtcatg 600
tgcaaatatt ttcttcgggt agactgcgcg agcatgcagc aactctgccc gccctatata 660
actagtgtct catcttcttg tcattcttgt ctctttacca acacaccggg actctttcac 720
catgggtaag taaaagacac acgtatcgcg accccgactc aactccaaca tggacgcaga 780
cctgtacgga tacaatatgg ctcgagataa tgtcggacag tcgggtgcga cgatctacag 840
actctatgga aaacccgagc cccccgagct ctttcttaaa cacggcaagg gtatcggttg 900
aaacgacgta accgacgaaa ggttcggtct gaattggcta actgagttca tgccactccc 960
caccatcaag cacttcattc gtacacctga cgatgcgtgg ctgttgacta cggccatccc 1020
cggttaagacc gcttttcagg tcctcgagga ataccctgac tcggggcaga acatcgtgga 1080
tgcttggctg gtttttttgc gccgccttca ctctattccg gtcgtcaact gccctttcaa 1140
cagcgaccgt gtttttaggc ttgccaggc acaatcacgc atgaacaatg gtctagtcca 1200
cgctagtgat ttcatgacg agcgcaacgg ctggccgggtg gaacaagtgt ggaaagagat 1260
gcataagctc ctgccccttt cccagactc gtgcgtgacg catgggtgatt tctcgcttga 1320
caacctcatc tttgacgagg gcaagctcat tggctgcatt gacgtcggcc gggttggcat 1380
cgccgatcgc taccagatc ttgcgacct ctggaattgt ctgggcgagt tctcccctag 1440
cctccagaag cgtctgttcc agaaatatgg gatcg 1475

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SEQ ID NO: 99      moltype = DNA length = 1621
FEATURE           Location/Qualifiers
source            1..1621
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 99
tattctggta cgagccatcc caaaaatgga tcatgacagc tgccaagagc caggactata 60
agattgagat ctacagcagc gatgacctca agtcttgaa gctcgaaagc gcctttgcga 120
acgaaggctt tcttggtcat cagtatgagt gccccggcct catcgagggt cctaccgagc 180
aagatcctag caagagctac tgggtgatgt tcatctctat caatccggga gcaccggcg 240
gcggctcctt caaccagtat ttgtgggtt cgttcaacgg aacgcacttt gaagcttttg 300
ataatcagtc aagagtcgtc gactttggca aagactacta cgccctccag accttcttta 360
acacggatcc aacgtacggt tcggcactgg gcacgcgctg ggctcgaaac tgggagtata 420
gcgcctttgt cccactaac ccttggcgct cgtcgatgct gttagtgcgt aagttctcgc 480

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tgaacaccga gtaccaggca aatccggaga cagaactaat caaccttaag gcagagccta 540
tcttgaacat ctctaacgcc ggaccctggg ctagggttcgc caccaacacc acgctcacta 600
aggccaatag ttacaatgtc gatctaagca attcaacggg cacactggaa ttcgagctcg 660
tttacgcagt caacaccacc cagacgatct cgaatccgt ctttcagat ttgtcgctgt 720
ggttcaaggg ccttcaggac cgggaagagt acctccgcat gggttttgaa gtctcagcgt 780
catcgttcct cctcgaccgc ggggaattcta aggtaaaagt cgtgaaggaa aacccctact 840
ttaccaatcg tatgtctgtg aacaatcagc cattcaaatc ggagaacgac ctgtcctact 900
acaagggtta cggcctctc gaccaaaca tcttcgagtt gtactttaac gacgggtgatg 960
ttgtgtcgac aaacacttat ttcatgacaa cgggaaacgc tctcggtacc gtcaatatga 1020
ctacgggtgt cgacaatctg ttttacattg acaaattcca ggtgcgagaa gtcaagtaat 1080
gggtttcacc actgtttgct cccctttgtt ggagcgtccg tcgttctgat ggcgctctga 1140
gccccaaata tcccttcttc gcccgagagc aagcctcgga gtgtgcttgc aatctacttt 1200
tctactaaa gttatcttct tgtggcttct gaatttgaat agaactgtgt cagacttgtt 1260
agctaaattt gttgatgaga gatacttgac agaactccgt tgatgaactc atctgtgtga 1320
ccaccgcctg gtgatggaat ctgcgcggga agcctgtgtg ttactgactc tcaagcaatt 1380
cagtagataa attgtataa catcattgca aggggtgcta gcaatcaatt gtccaatacc 1440
aaacggcgcc aagtgcaagg gggcgataga gagtttctgc gaggaaggga aggaatccgt 1500
atgcctgaaa acttttgtac aagtgtcttt gtaaggcgga ggtctgctct gacatctctc 1560
tgagatctga gatctgatac ctgcaggata acttcgtata atgtatgcta tacgaagtta 1620
t 1621

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SEQ ID NO: 100      moltype = DNA length = 1982
FEATURE             Location/Qualifiers
source              1..1982
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 100
ataacttcgt ataagtatg ctatacgaag ttatcctgca ggagctcaga ccttgcattgt 60
tagccttgac cttgcctcta tgaccactaa ttgcacaaag attgcacggt accattgtca 120
tcttgccatg tatccaatgt ttctctgttg ccatagtacg tagtagatac atcgatcatc 180
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cc 1982

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FEATURE             Location/Qualifiers
source              1..500
                    mol_type = other DNA
                    organism = Moniliella pollinis

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agcaacagtg gtgaacacca                                     500

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1. A genetically engineered yeast cell capable of producing xylitol, the engineered yeast cell comprising:

an exogenous polynucleotide sequence encoding a xylulokinase (XKS) enzyme comprising a sequence at least 80% identical to at least one of SEQ ID NOs:8 and 20.

2. The yeast cell of claim 1, wherein the yeast cell is an osmotolerant yeast cell.

3. The yeast cell of claim 1, wherein the yeast cell is a cell of the subphylum Ustilaginomycotina.

4. The yeast cell of claim 1, wherein the yeast cell is selected from the group consisting of *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*, *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Moniliella*, *Ustilaginomyces*, *Trichosporon*, *Yarrowia lipolytica*, *Penicillium*, *Torula*, *Pichia*, *Candida*, *Candida magnoliae*, and *Aureobasidium*.

5. The yeast cell of claim 1, wherein the yeast cell is a yeast cell of the *Moniliella* genus.

6. (canceled)

7. The yeast cell of claim 5, wherein the cell is a *Moniliella pollinis* cell.

8. The yeast cell of claim 1, wherein the yeast cell is capable of producing xylitol at a titer of at least 1.0 g/L when used in a fermentation process in the presence of dextrose for 96 hours.

9. The yeast cell of claim 1, wherein the exogenous polynucleotide sequence is integrated into the genome of the yeast cell at a locus selected from the ER1 locus, the ER3 locus, the PDC1 locus, the pyrF locus, the TRP3 locus, the gpdIIA locus, and the gpdIIB locus.

10. The yeast cell of claim 1, wherein the exogenous polynucleotide sequence is operably linked to a heterologous or artificial promoter selected from the group consisting of pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:61), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:52), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:54), translational elongation factor 1 promoter (TEFp; SEQ ID NO:55), modified TEFp (SEQ ID NO:53), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:56), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:57), enolase promoter (ENO1p; SEQ ID NO:58), asparagine synthetase promoter (ASNSp;

SEQ ID NO:59), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:60), and RPL16B (SEQ ID NO:62).

11. (canceled)

12. (canceled)

13. The yeast cell of claim 1, wherein the yeast cell additionally comprises

an exogenous polynucleotide sequence encoding a xylitol dehydrogenase (XDH) enzyme comprising a sequence at least 80% identical to at least one of SEQ ID NOs:1, 13, and 14.

14. (canceled)

15. (canceled)

16. The yeast cell of claim 1, wherein the sugar phosphatase enzyme has a sequence at least 85% identical to at least one of SEQ ID NOs:8 and 20.

17. The yeast cell of claim 1, wherein the sugar phosphatase enzyme has a sequence at least 90% identical to at least one of SEQ ID NOs:8 and 20.

18. The yeast cell of claim 13, wherein the XDH enzyme has a sequence at least 85% identical to at least one of SEQ ID NOs:1, 13, and 14.

19. The yeast cell of claim 13, wherein the XDH enzyme has a sequence at least 90% identical to at least one of SEQ ID NOs:1, 13, and 14.

20. A method for producing xylitol, the method comprising:

contacting a substrate comprising dextrose with the engineered yeast cell of claim 1, wherein fermentation of the substrate by the engineered yeast produces xylitol.

21. (canceled)

22. (canceled)

23. The method of claim 20, wherein the fermentation temperature is at or between 25° C. to 45° C.

24. The method of claim 20, wherein xylitol production is at least 1.0 g/L when the fermentation is run at 35° C. for 96 hours.

25. (canceled)

26. (canceled)

27. The method claim 20, wherein the volumetric oxygen uptake rate (OUR) is between 0.5 to 40 mmol O₂/(L·h).

28. (canceled)

29. (canceled)

30. (canceled)

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